

BSP Build Guide

for Windows[®] CE 5.0

VRender[™] 3D

Build Guide

MDVR3511FBBUM-WC50

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VR3511F

*Platform Builder for Microsoft Windows[®] CE 5.0
for VR3511F*

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WinCE.NET BSP Build Guide for VR3511F

1. Creating a New Platform

1.1 Installing BSP

Create **VR3511F** folder under **WINCE500\PLATFORM** folder

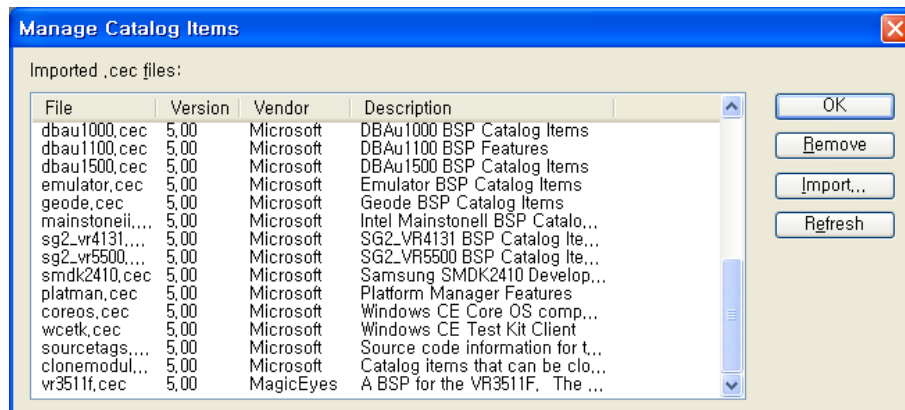
Uncompress BSP File in CD to **WINCE500\PLATFORM\ VR3511F** Folder

CAUTION

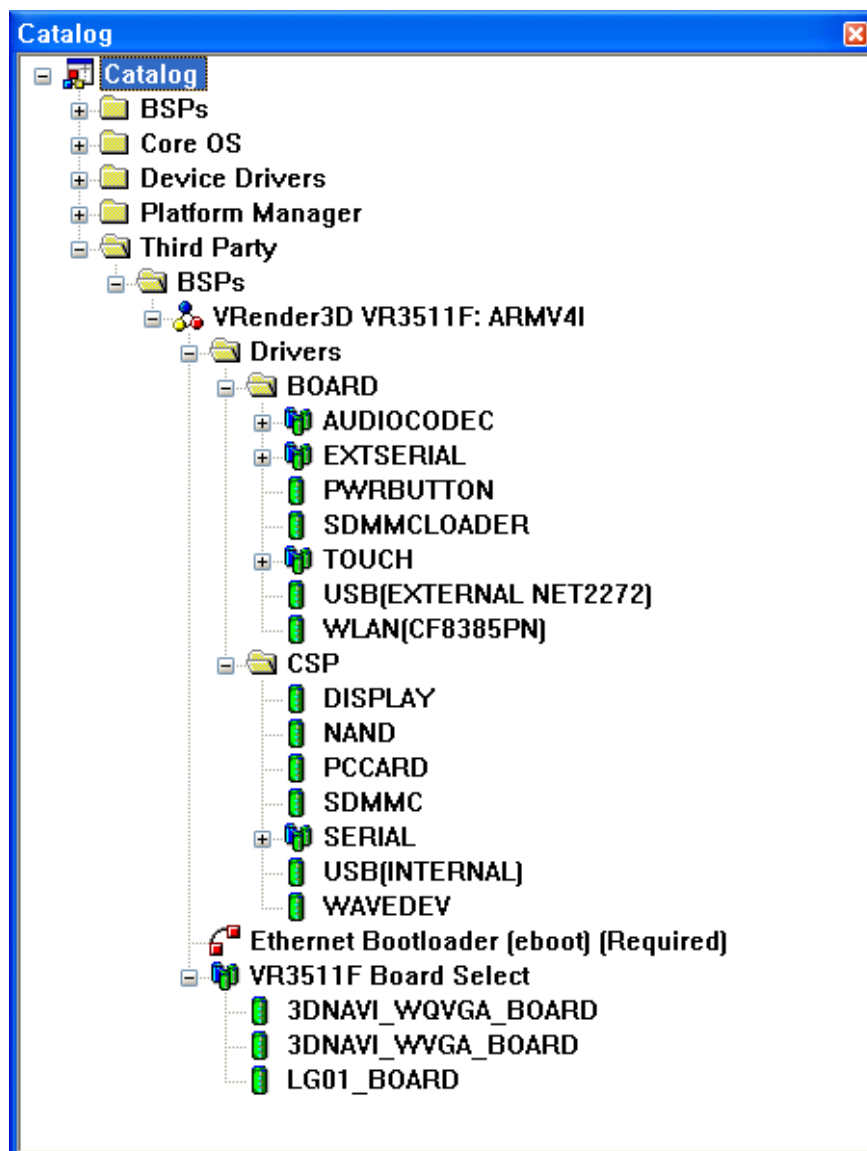
- If you have previous version of the BSP, remove it completely before install the new version of the BSP..
- Close platform builder and editor application before you install the new version of BSP.

1.2 Catalog Registration of BSP

1. Run Platform Builder
2. Select **Manage Catalog Features** in **File Menu**..
3. Press **Import** button and select **VR3511F.cec** in VR3511F folder.

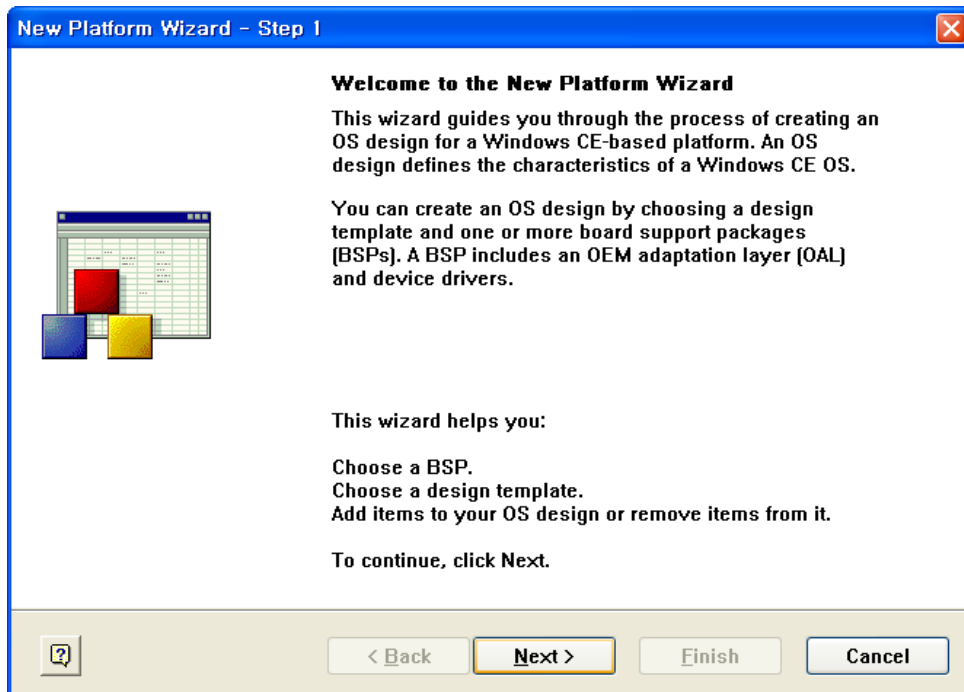


4. Register imported .cec file by pressing OK.
5. Now VR3511F BSP is added in catalog list.

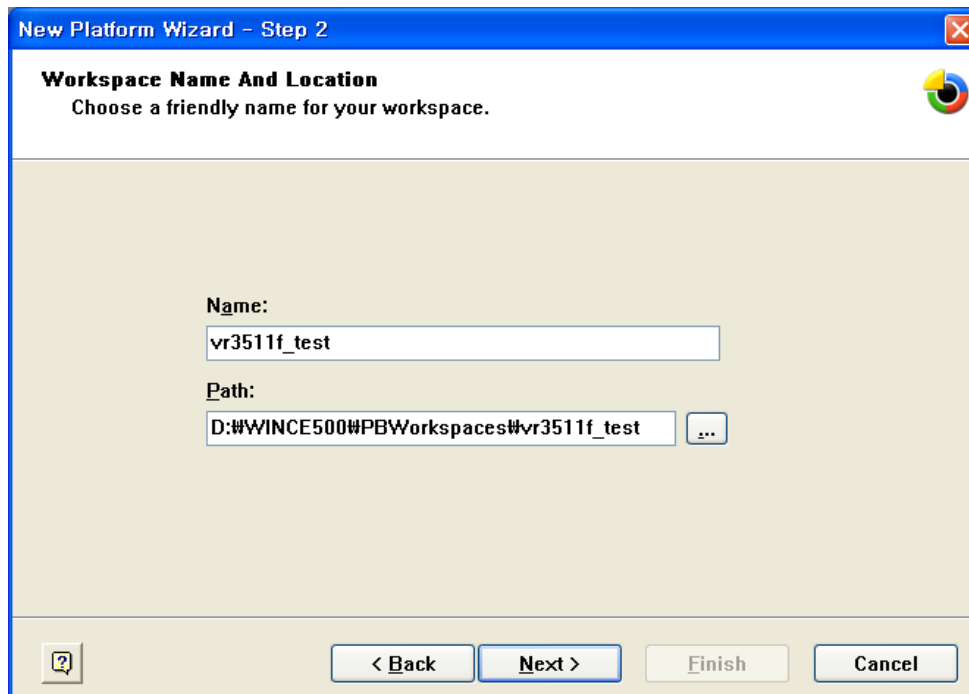


1.3 Creating New VR3511F Platform

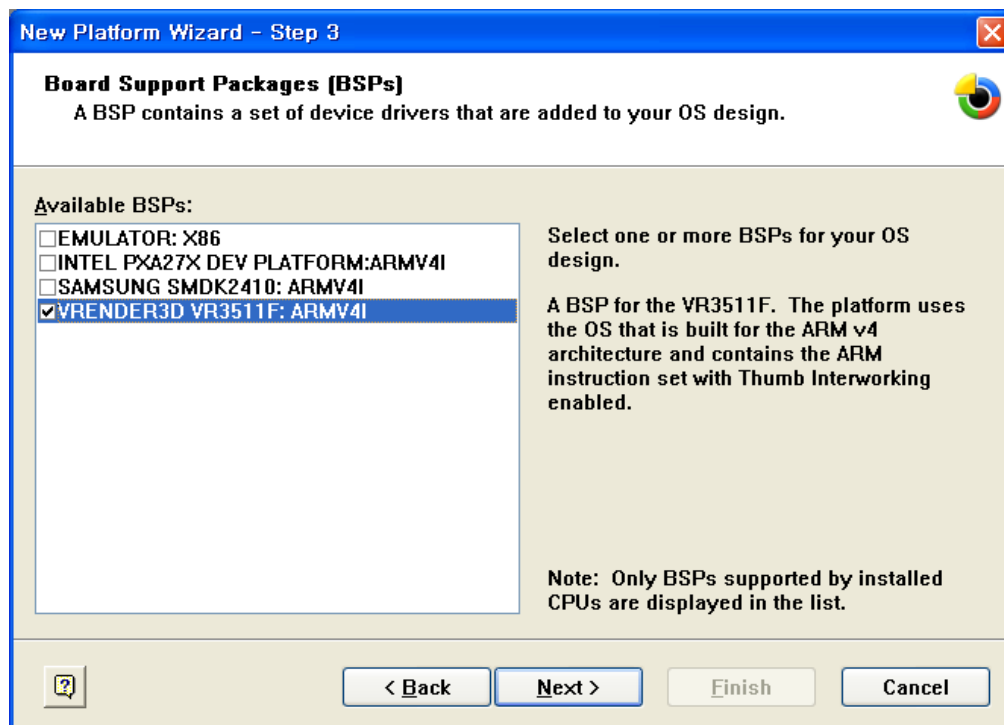
1. Go to **File -> New Platform** in Platform Builder menu.



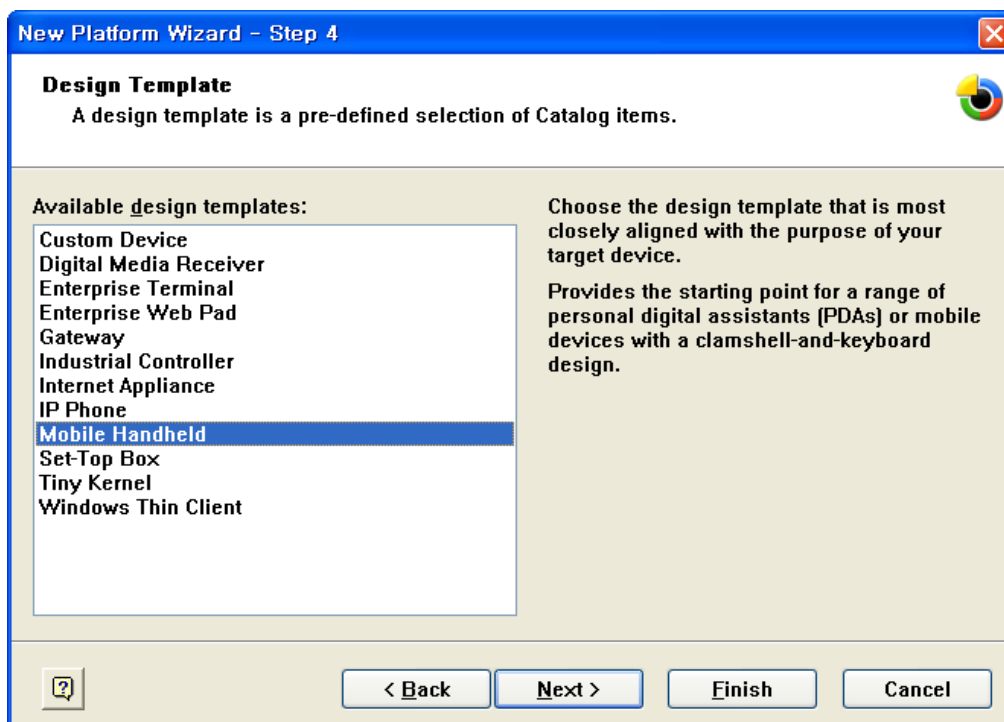
2. Press **Next** and choose a name for new workplace.



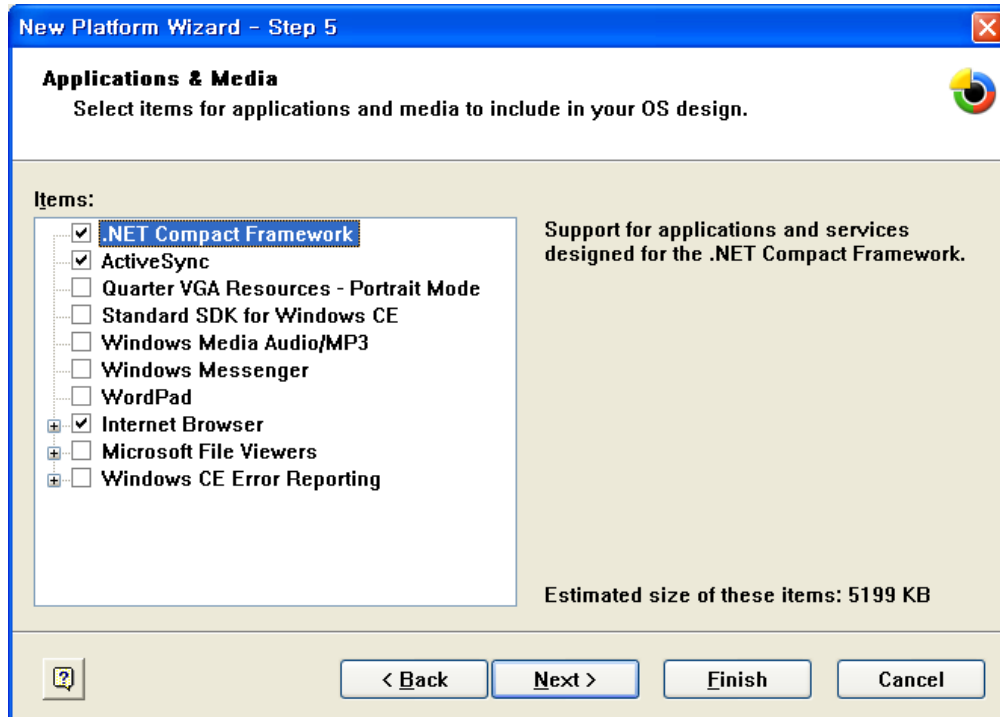
3. Press **Next** and check **VR3511F:ARMV4I** in New Platform Wizard.



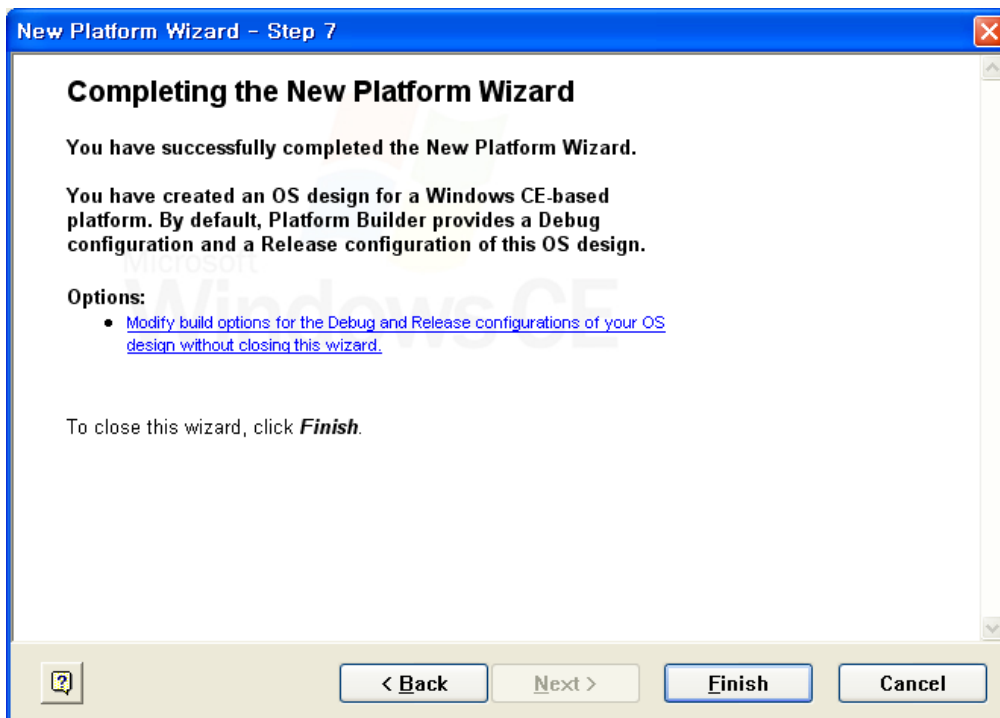
4. Select appropriate configuration template.



5. Press **Next >** and check features needed. (Features shown in this window can be added or removed in Catalog window.)



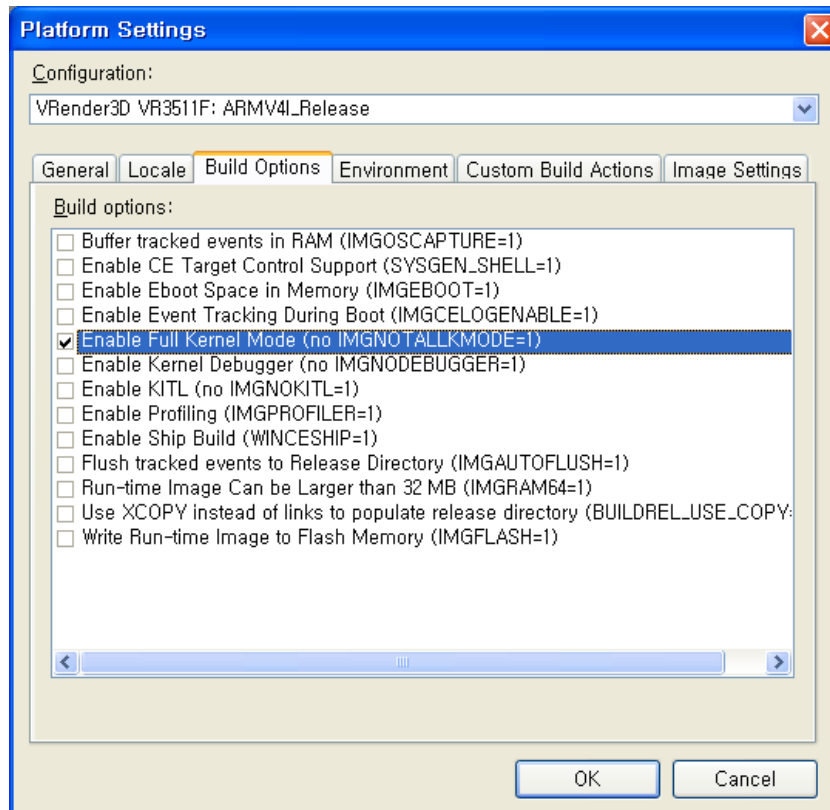
6. Complete New Platform creation by pressing **Finish** button.



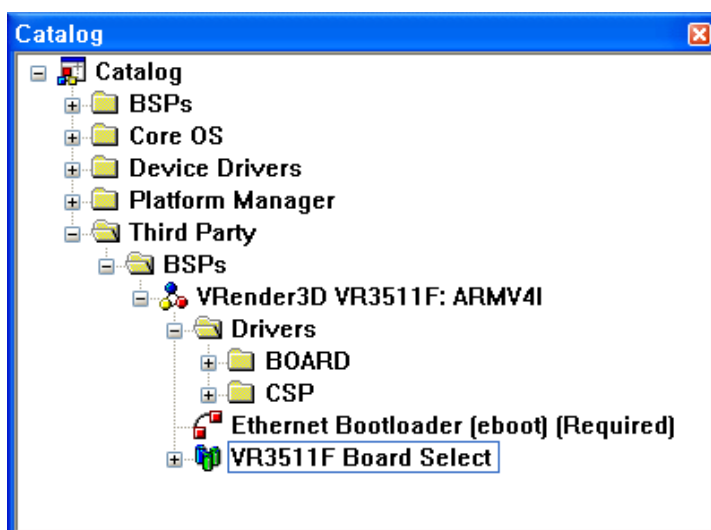
2. Platform Configuration

2.1 Configuration Settings for Development Board

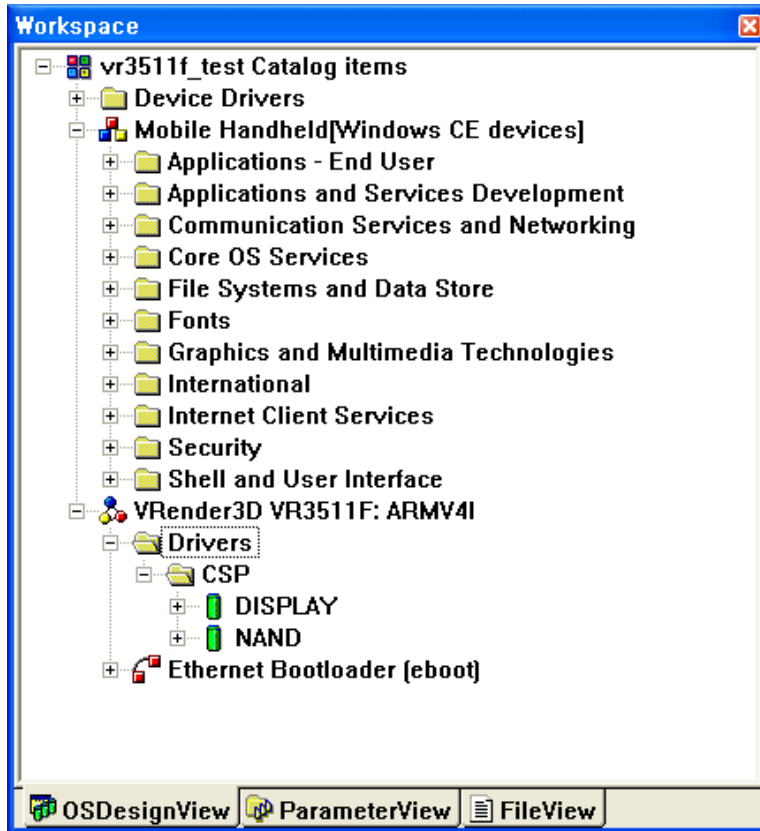
1. Select **Platform** -> **Settings** in Platform Builder Menu.
2. Check **Enable Full Kernel Mode** in the **Build Options** list.



Add appropriate board configuration from Catalog Window.



2.2 Adding Basic Components



Add necessary components according to the driver guides in this document.

(Errors can be occurred during build process when components required by each driver are not added and configured correctly.)

For configuring the drivers and components, you must refer to each driver guide.

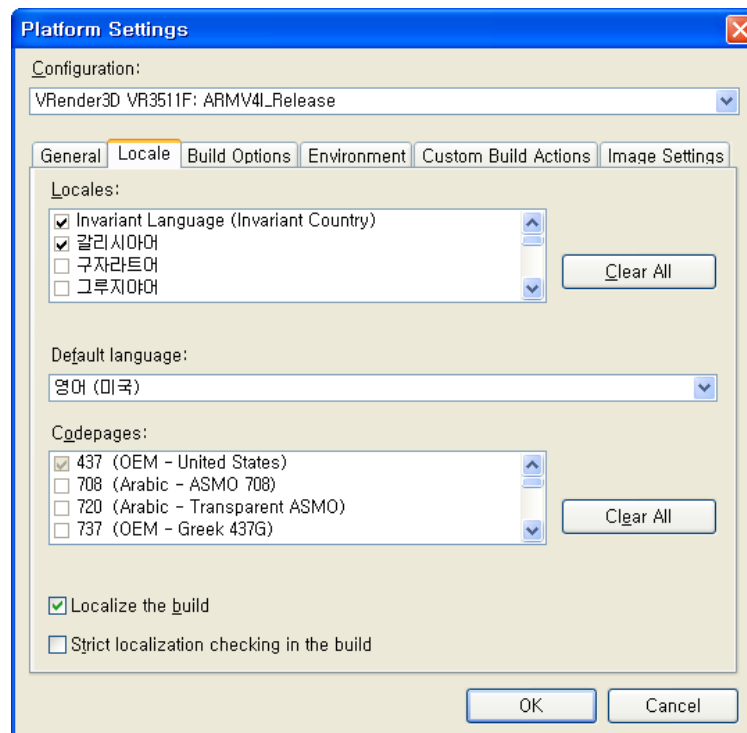
You can reduce image of the Windows CE OS by changing default language to English rather than Asian language such as Korean, Chinese, or Japanese. To change default language:

Select **Platform** -> **Settings** Menu.

Clear all locales by pressing **Clear All** button of **Locales**.

Select English from **Locales** list.

Select **Default Language** as English.



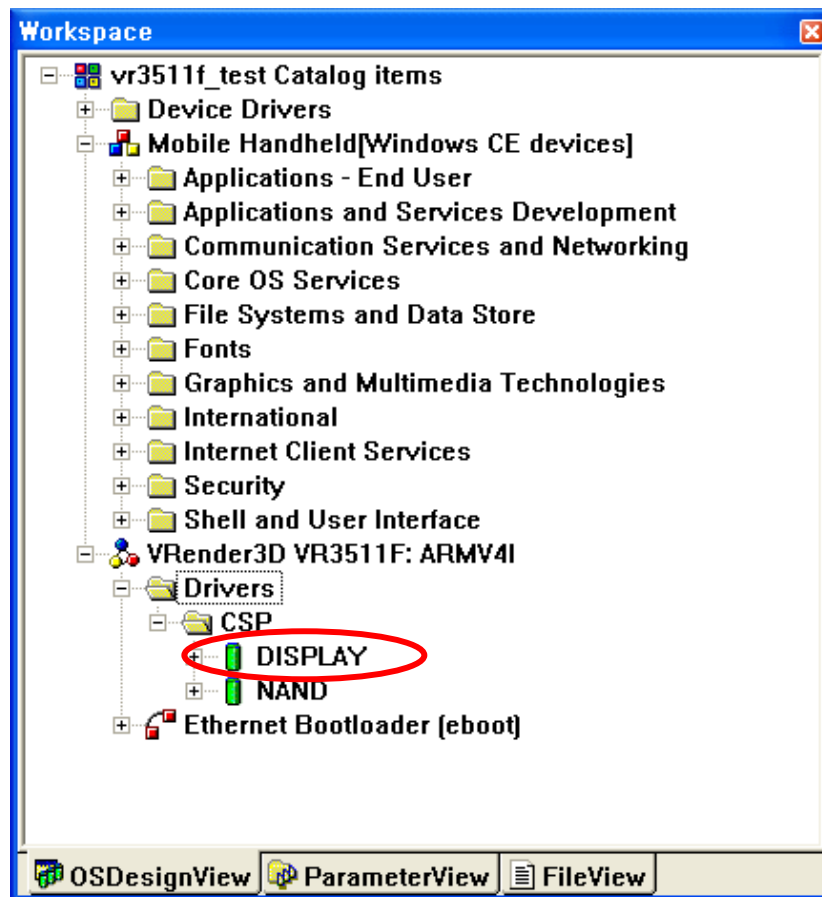
This is a sample of locale setting of Platform Builder Workspace than runs on Korean Windows XP.

3. Display Driver Guide

3.1 Adding Display Driver

3.1.0 Adding Display device driver

Add **DISPLAY** component in **Third Party** → **BSPs** → **VRender3D VR3511F: ARMV4I** → **Drivers** → **CSP** → **DISPLAY** of the catalog. (Click this item with right button of the mouse and select 'Add to OS Design' menu.)



4. USB Host Driver Guide

4.1 Using USB Mouse

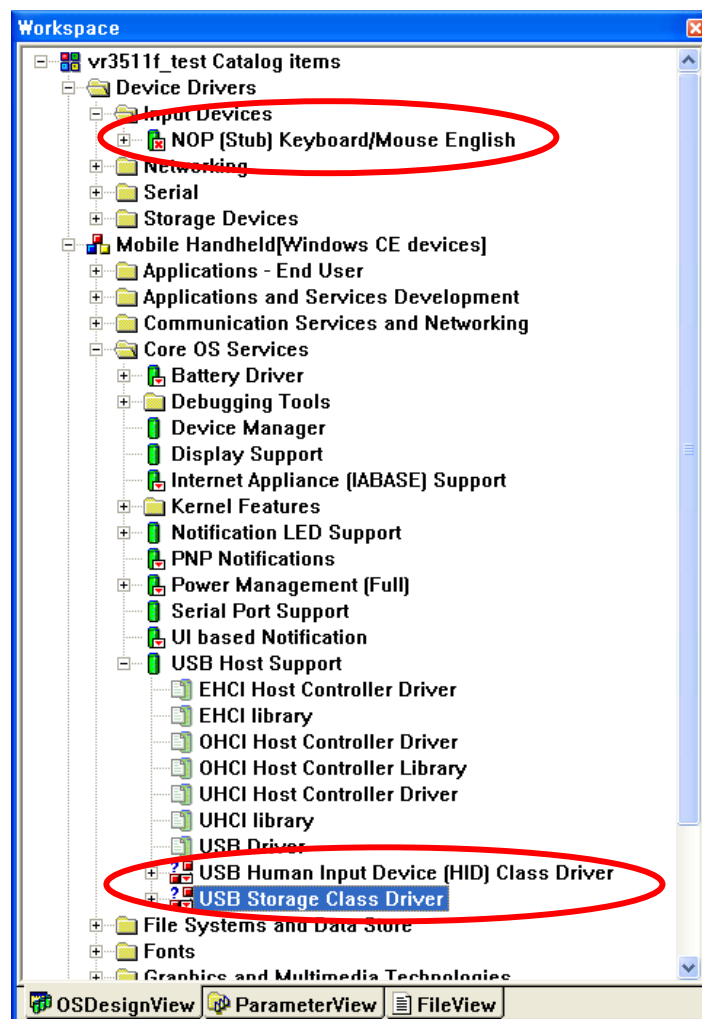
To use USB mouse, add **Core OS → Windows CE devices → Core OS Services → USB Host Support → USB Human Input Device (HID) Class Driver → USB HID Keyboard and Mouse** component.

4.2 Using USB Keyboard

To use USB keyboard, add **USB HID Keyboard and Mouse** component and **Device Drivers -> Input Devices -> Keyboard/Mouse -> NOP(Stub) Keyboard/Mouse English** component.

4.3 Using USB Storage Device

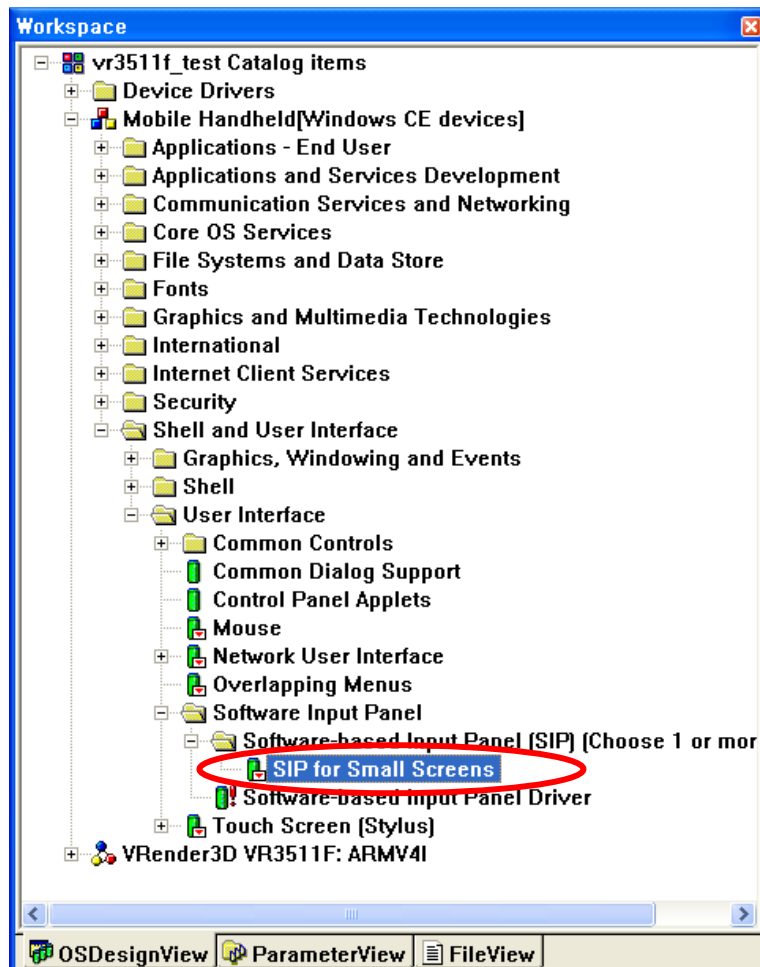
To use USB storage devices, add **Core OS → Windows CE Devices → Core OS Services → USB Host Support → USB Storage Class Driver** component. (See next picture)



4.4 Using Virtual Keyboard

To use Virtual Keyboard instead of USB keyboard, add **Core OS** → **Windows CE Devices** → **Shell and User Interface** → **User Interface** → **Software Input Panel** → **Software-based Input Panel [SIP] [Choose 1 or more]** → **SIP for Large/Small Screens** component.

Windows CE supports two types of virtual keyboard, small and large. You can select either of them according to the screen configuration of your system.



5. USB Function Driver Guide

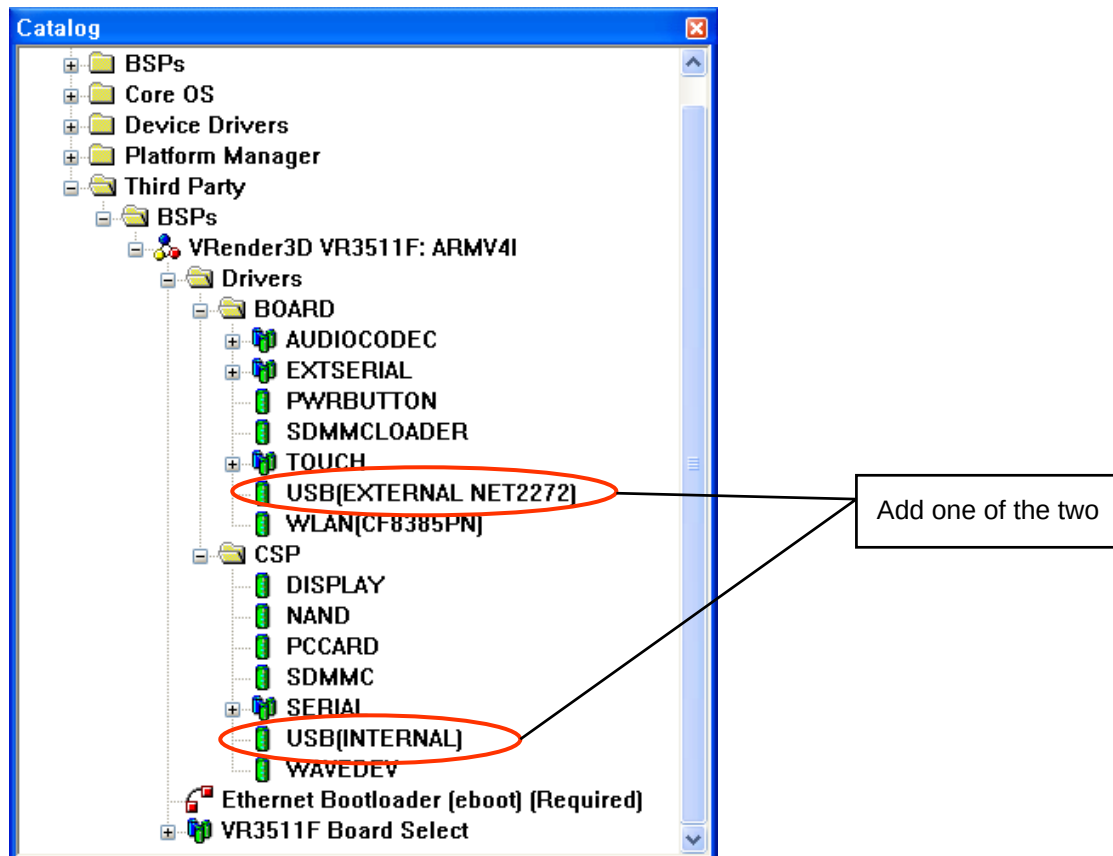
5.1 Adding USB Function Driver Component

To use USB Function, one of two components below is required. If you add both of them, USB(INTERNAL) component has higher priority and USB(EXTERNAL NET2272) component is ignored.

Add **Third Party** → **BSPs** → **VRender3D VR3511F:ARMV4I** → **Drivers** → **CSP** → **USB(INTERNAL)** component. (Click this item with right button of the mouse and select 'Add to Platform' menu.)

or

Add **Third Party** → **BSPs** → **VRender3D VR3511F:ARMV4I** → **Drivers** → **BOARD** → **USB(EXTERNAL NET2272)** component. (Click this item with right button of the mouse and select 'Add to OS Design' menu.)



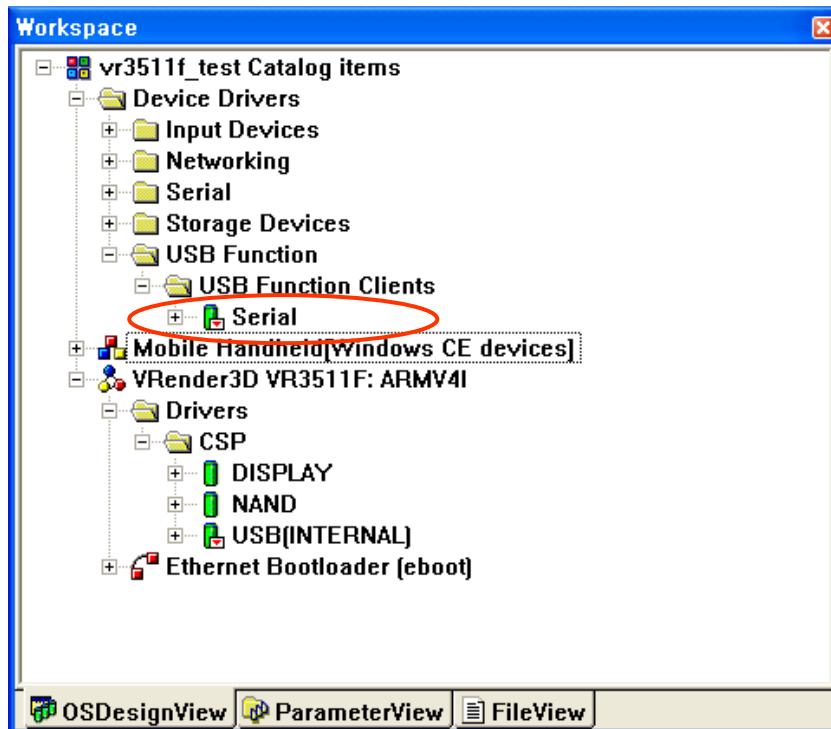
NOTE : You can add only one USB Function client at once. For example, if you add Serial component, Mass Storage component can not be added.

5.2 Using ActiveSync with USB Connection

In advance, make sure one of the USB function drivers above is added.

Add **Device Drivers** → **USB Function** → **USB Function Clients** → **Serial** component. (Click this item with right button of the mouse and select 'Add to Platform' menu.)

NOTE : You cannot use ActiveSync with USB Mass Storage simultaneously. Make sure USB Mass Storage is not added from catalog.

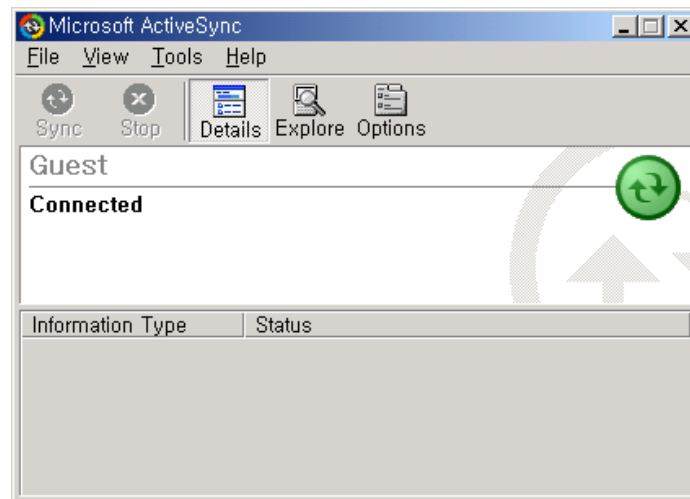


To test ActiveSync on DTK board, make sure things below are done.

- USB connection cable is prepared. (One side is A type, and the other side B type)
- ActiveSync application is installed on host PC.
- When you set configuration of new project with **New Platform Wizard**, Check **ActiveSync** feature.
- Build image with above configuration and other settings shown on this BSP guide.

After Windows CE image is built. Connect USB cable and load Windows CE image on DTK board. You can see the ActiveSync connection with host PC.

When some settings are completed, ActiveSync window will be shown like this:

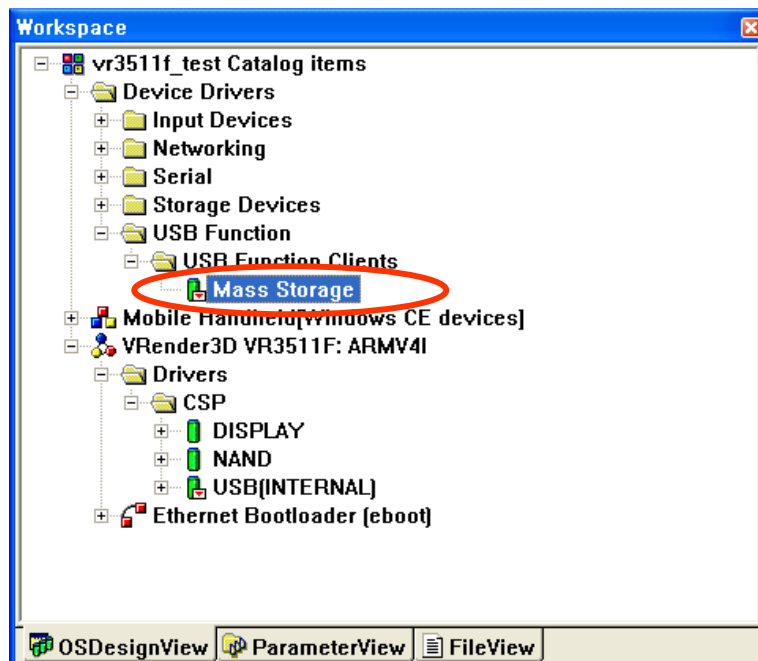


5.3 Using USB Mass Storage

In advance, make sure one of the USB function drivers above is added.

Add **Device Drivers** → **USB Function** → **USB Function Clients** → **Mass Storage** component.

(Click this item with right button of the mouse and select 'Add to Platform' menu.)



When USB 2.0 mass storage driver is initialized correctly, connect USB cable with host system. (e.g. desktop PC)

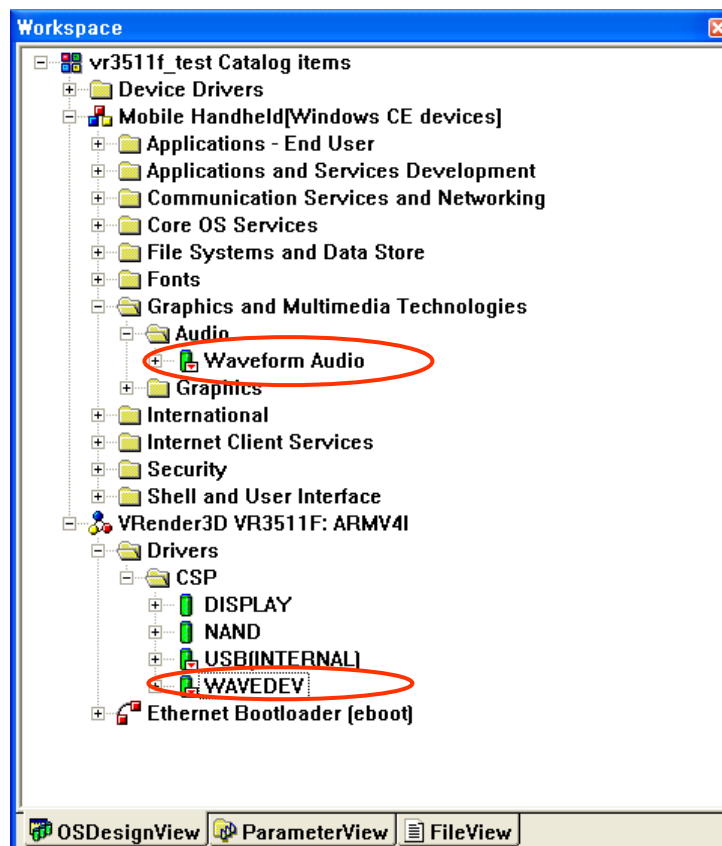
- On the target board, use left one of the two USB device connectors.

After connection is established, VR3511F based system is mounted as a removable disk on the host side.

6. AC97 Driver Guide

6.1 Adding Required Components

- Add **Third Party** → **BSPs** → **VRender3D VR3511F:ARMV4I** → **Drivers** → **CSP** → **WAVEDEV** component. (Click this item with right button of the mouse and select 'Add to OS Design' menu.)
- Add **Core OS** → **Windows CE Devices** → **Graphics and Multimedia Technologies** → **Audio** → **Waveform Audio** component for basic waveform audio play. (Click this item with right button of the mouse and select 'Add to OS Design' menu.)
- Add one of the **AUDIOCODEC** components in **Third Party** → **BSPs** → **VRender3D VR3511F:ARMV4I** → **Drivers** → **BOARD** → **AUDIOCODEC**. If you use a general AC97 device, add **AUDIOCODEC(AC97)**. If you use a WM9712 device, add **AUDIOCODEC(WM9712)**.



6.2 Adding Optional Components

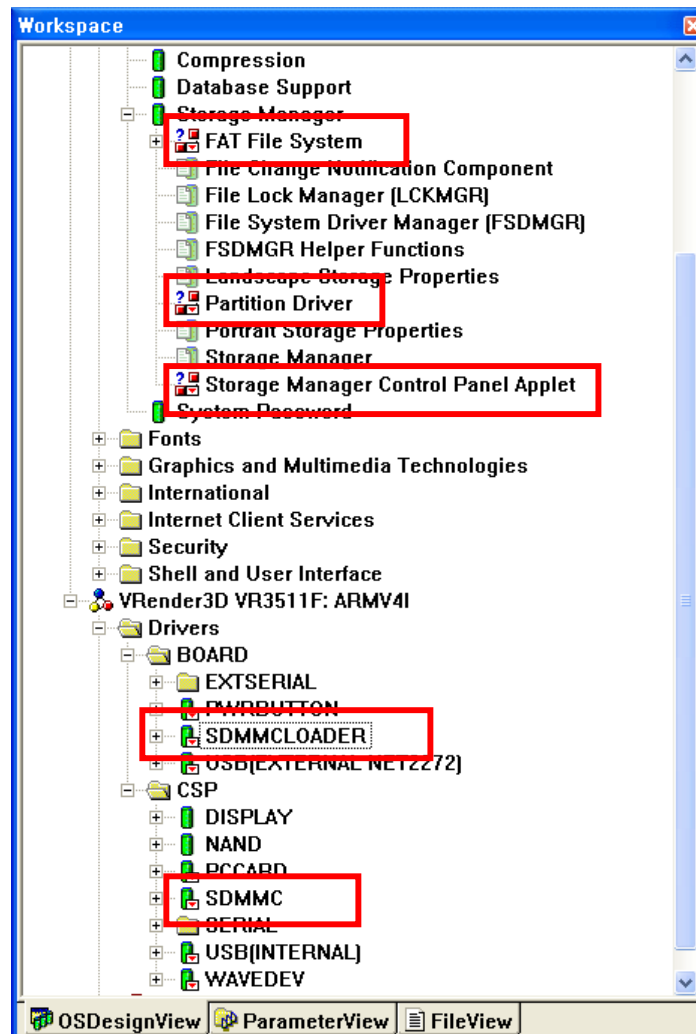
To play other format of audio files, appropriate components should be added.

7. SD/MMC Storage Driver Guide

7.1 Adding SD/MMC Storage Device Driver

Add Third Party → BSPs → VRender3D VR3511F:ARMV4I → Drivers → CSP → SDMMC component. (Click this item with right button of the mouse and select 'Add to Platform' menu.)

Add Third Party → BSPs → VRender3D VR3511F:ARMV4I → Drivers → BOARD → SDMMCLoader component.



7.2 Adding Required Components

- Add below components in Core OS → Windows CE Devices → File System and Data Store → Storage Manager. (See above picture)
 - FAT File System
 - Partition Driver
 - Storage Manager Control Panel Applet

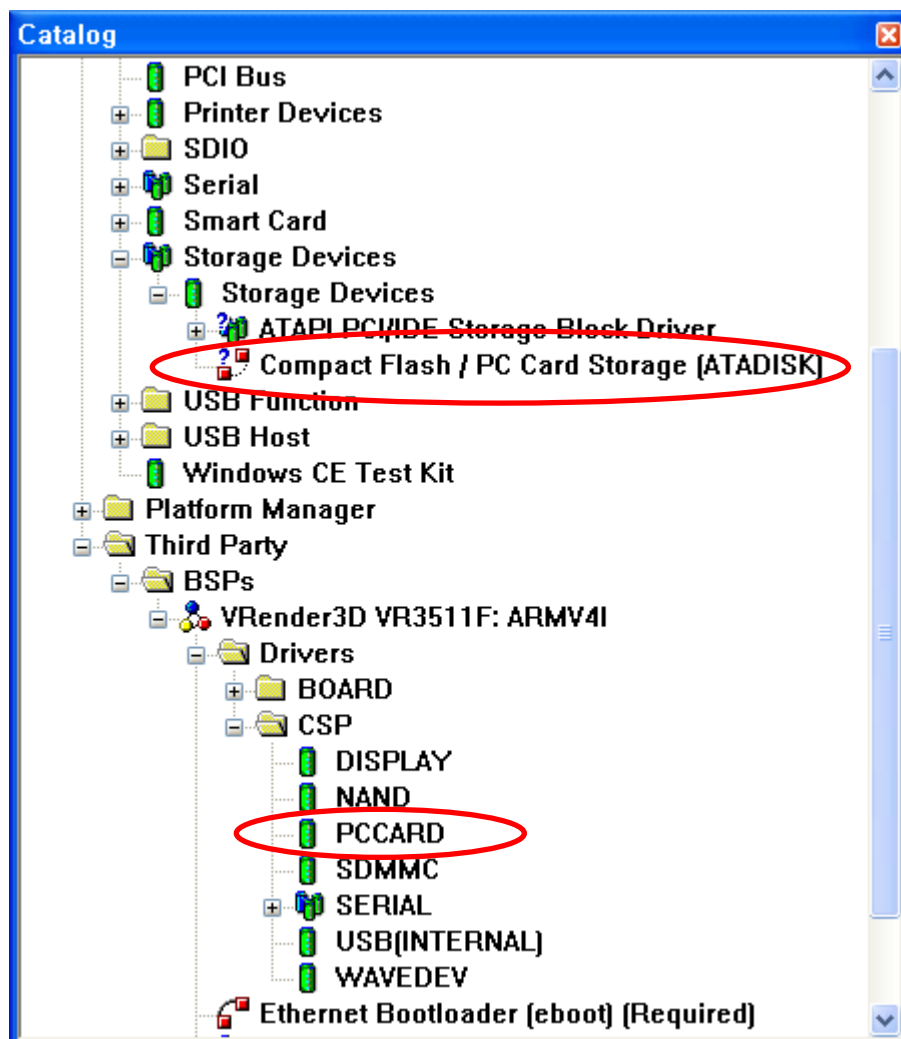
8. CompactFlash Storage Driver Guide

8.1 Adding CompactFlash Storage Device Driver

CompactFlash storage driver runs on the CompactFlash/PCMCIA bus driver layer. So you must include both CompactFlash storage and CompactFlash/PCMCIA bus driver.

Add **Third Party** → **BSPs** → **VRRender3D VR3511F:ARMV4I** → **Drivers** → **CSP** → **PCCARD** component. (This is the CompactFlash / PCMCIA host driver. CF storage driver runs on this host layer.)

Add **Device Drivers** → **Storage Devices** → **Storage Devices** → **CompactFlash / PC Card Storage (ATADISK)** component. (This is the CompactFlash storage driver.)



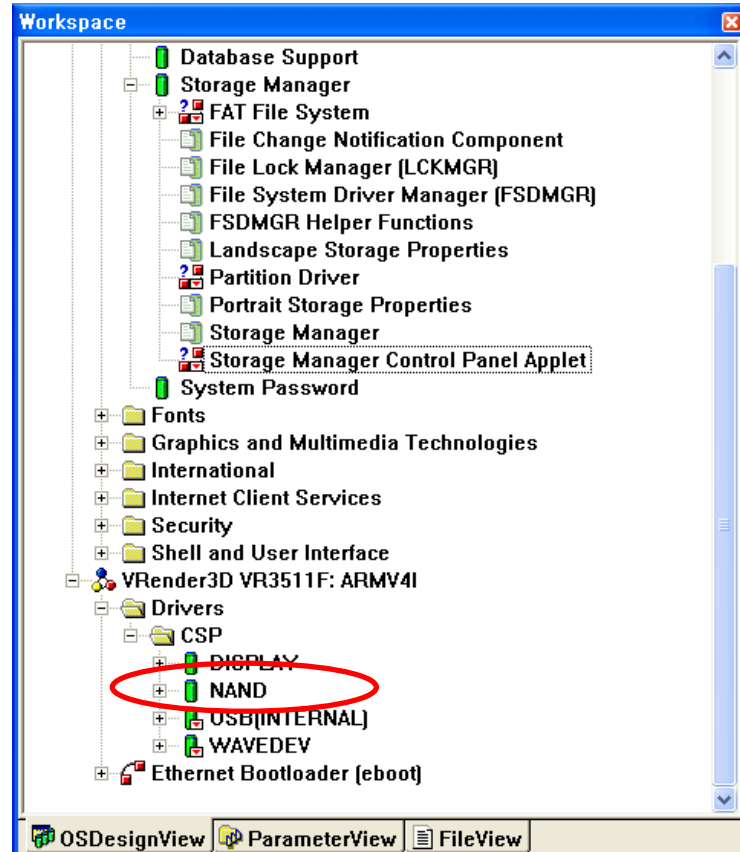
8.2 Adding Required Components

- Add FAT File System Component so that disk can be formatted as FAT file system. (See 'Required Components' of the **SD/MMC Storage Driver** section)

9. NAND Flash Storage Driver Guide

9.1 Adding NAND Flash Storage Device Driver

Add Third Party → BSPs → VRender3D VR3511F: ARMV4I → Drivers → CSP → NAND component.



9.2 Required Components

- Add FAT File System Component so that disk can be formatted as FAT file system. (See 'Required Components' of the **SD/MMC Storage Driver** section)

10. Serial Driver Guide

There are six serial drivers Support 6 Serial Ports. VR3511F can support 2 internal serial ports and four 16550 external serial ports.

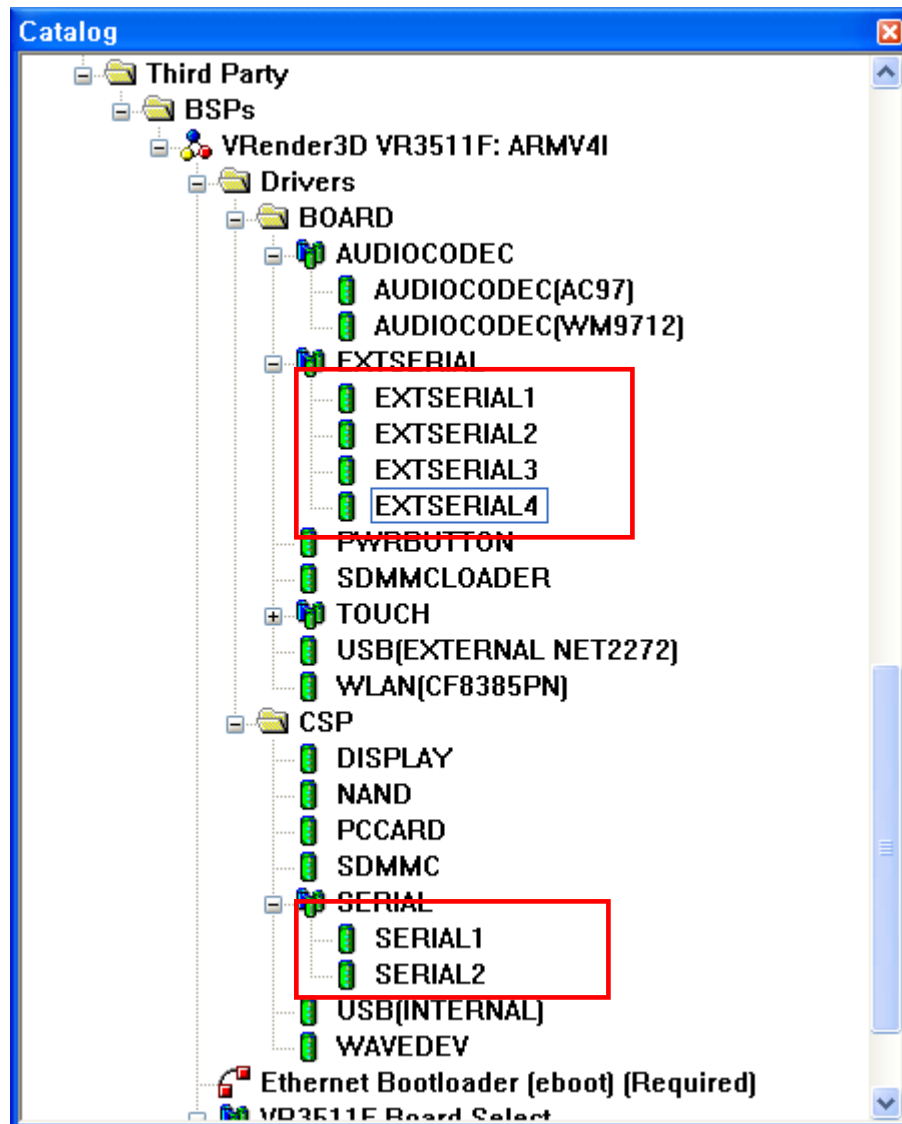
NOTE : If you use Serial Port1 or 2 for debugging. You cannot use that Serial Port for other purpose.

10.1 Adding Serial Driver on Chip

- A. Add Third Party → BSPs → VRender3D VR3511F:ARMV4I → Drivers → CSP → SERIAL → SERIAL#

10.2 Adding Serial Driver on Board

- B. Add Third Party → BSPs → VRender3D VR3511F:ARMV4I → Drivers → BOARD → EXTSERIAL → EXTSERIAL#



10.3 Running Serial Test Applications

- A. Open
\$(_WINCEROOT)\PLATFORM\VR3511F\SRC\APPS\SerialTest\CESerialTestClient\CESerialTestClient.vcw with Micorosoft eMbedded Visual C++ and build it. Then run 'CESerialTestClient.exe' in the target folder.
- B. Select Connect Port and choose Port name and Baud rate. and select OK.
- C. Open
\$(_WINCEROOT)\PLATFORM\VR3511F\SRC\APPS\SerialTest\PCSerialTest\SerialTest.dsw with Micorosoft Visual Studio and build it. Then run **SerialTest.exe** in you PC.
- D. Select **Connect** → **Port Connect**. choose Port number and Baud rate same as CESerialTestClient set. A value of Send Data Size is useless. A size of data to be send is fixed at 100KB.
- E. Click Start.
- F. If CESerialTestClient get perfect, CESerialTestClient show MessageBox.
- G. Select OK in the MessageBox then CESerialTestClient Send to PC SerialTest Application.
- H. If PC SerialTest Application get Perfect, Complete message will be shown on list control.

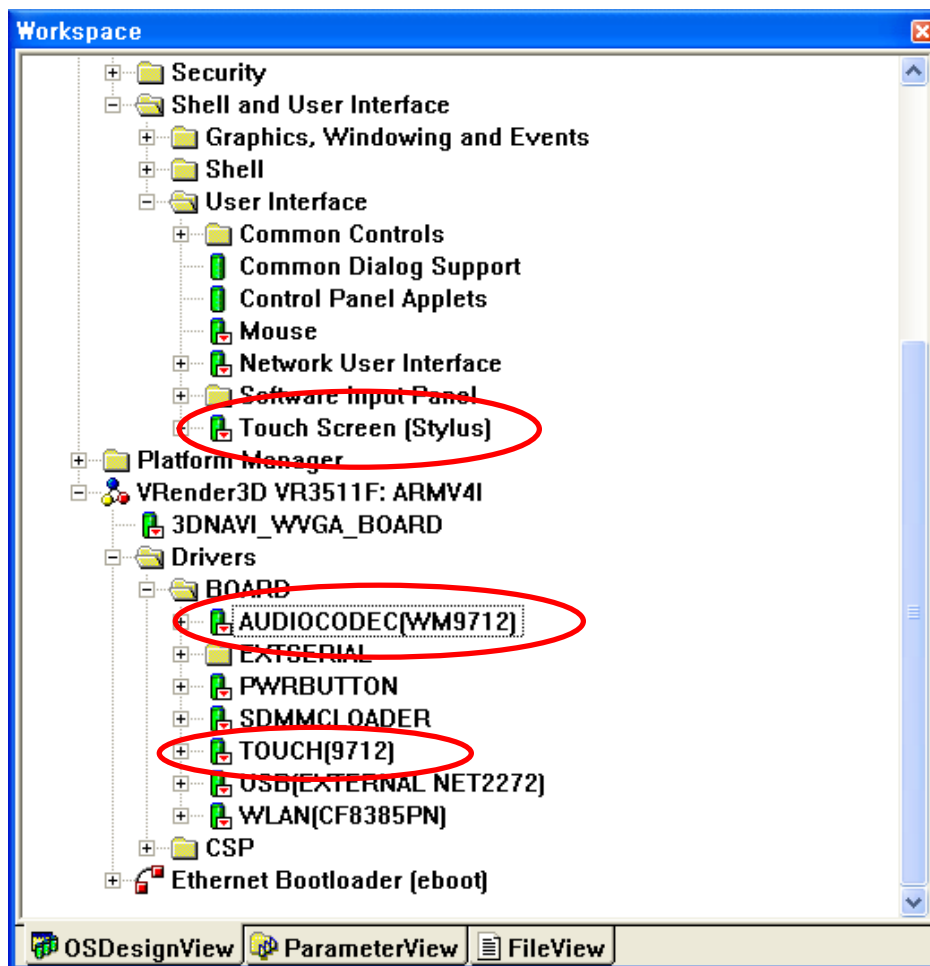
11. Touch Screen Driver Guide

11.1 Adding Required Components

Add Third Party → BSPs → VRender3D VR3511F:ARMV4I → Drivers → BOARD → TOUCH → TOUCH (xxx)

Add Core OS → Windows CE devices → Core OS Services → Shell and User Interface → User Interface → Touch Screen (Stylus) component.

In case of adding TOUCH(9712), add Third Party → BSPs → VRender3D VR3511F:ARMV4I → Drivers → BOARD → AUDIOCODEC → AUDIOCODEC(WM9712) component too.



12. DTK Board Selection

VR3511F supports two types of DTK board.

3DNAVI_WQVGA_BOARD: A board equipped with WQVGA LCD and targeting on 3D Navigations.

3DNAVI_WVGA_BOARD: A board equipped with WQVGA LCD and targeting on 3D Navigations.

13. Power Button Driver Guide

VR3511F DTK has Momentary Power On Button.

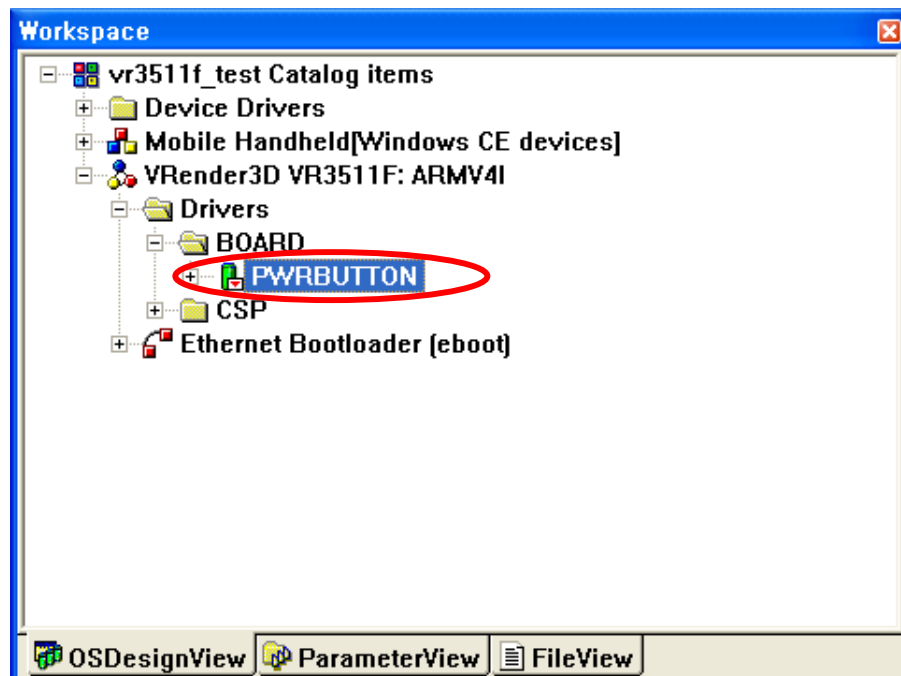
If you push the button for short time, the board enters suspend mode. It wakes up when you push the button once again.

If you push the button for long time, the power is off.

You can set the time by modifying a source code in BSP. If you want to get further information of it, refer to porting guide.

13.1 Adding Power Manager Driver

Add Third Party → BSPs → VRender3D VR3511F:ARMV4I → Drivers → BOARD → PWRBUTTON

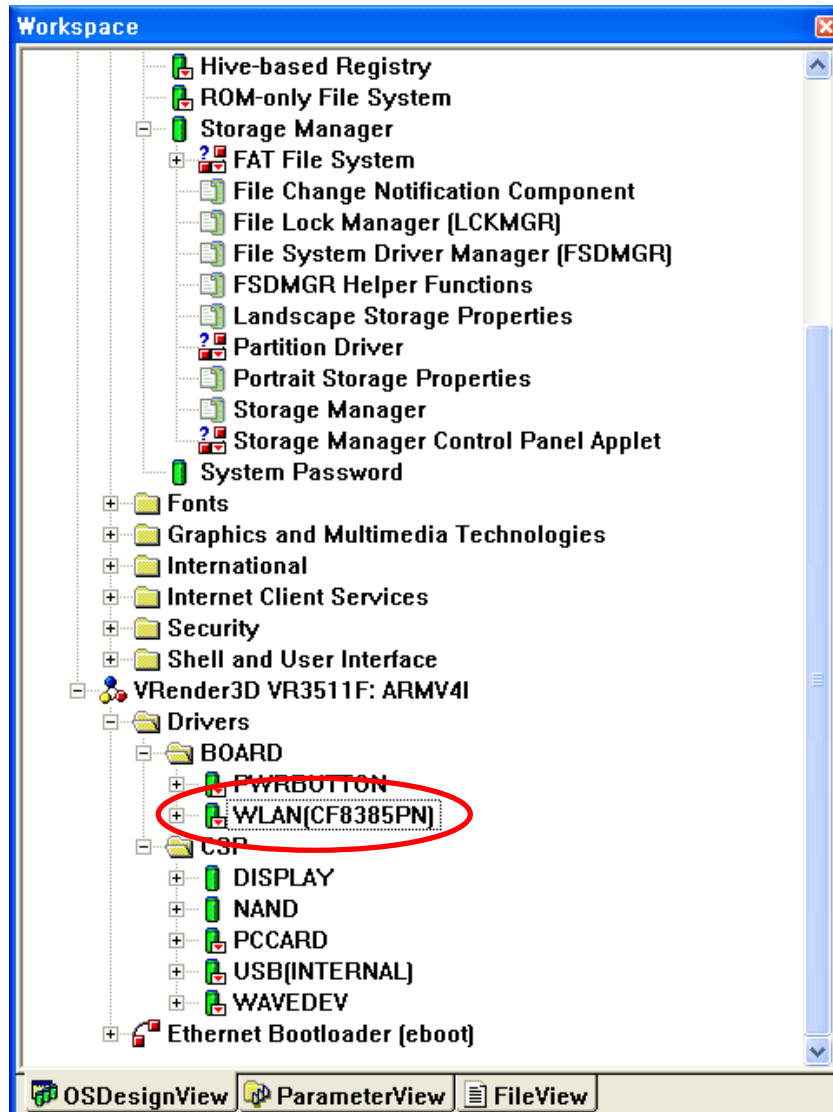


14. Wireless LAN Driver Guide

VR3511F support CFIO wireless LAN device which is CF8385PN.

14.1 Adding Wireless LAN Driver

Add Third Party → BSPs → VRender3D VR3511F:ARMV4I → Drivers → BOARD → WLAN(CF8385PN)



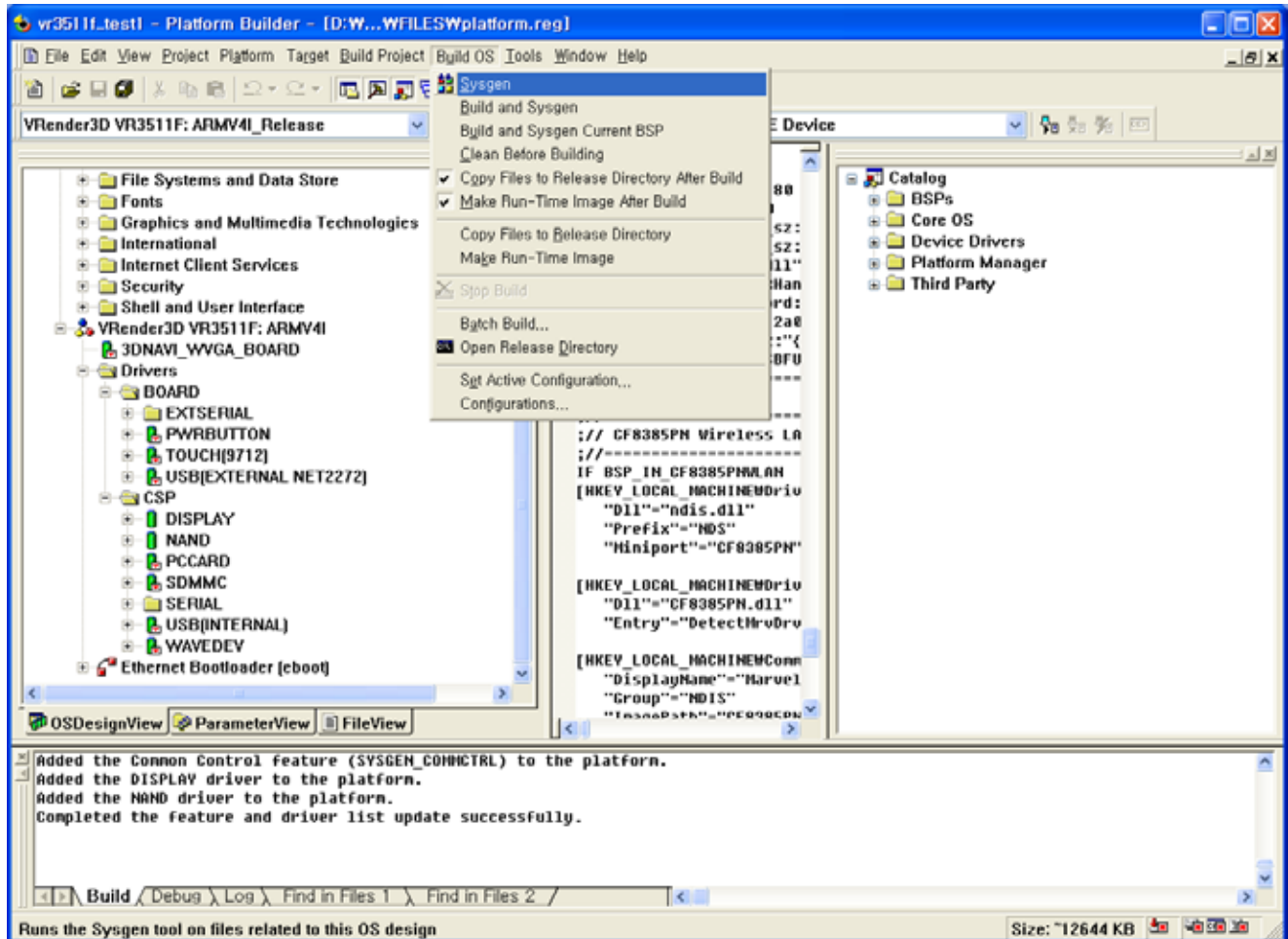
14.2 Adding a required driver

Add Compact Flash Storage Driver to use Wireless LAN properly. (See 'CompactFlash Storage Driver Guide'.)

15. Building WinCE OS

15.1 Building Entire OS

When setting is completed, start building by selecting **Build OS -> Sysgen**.



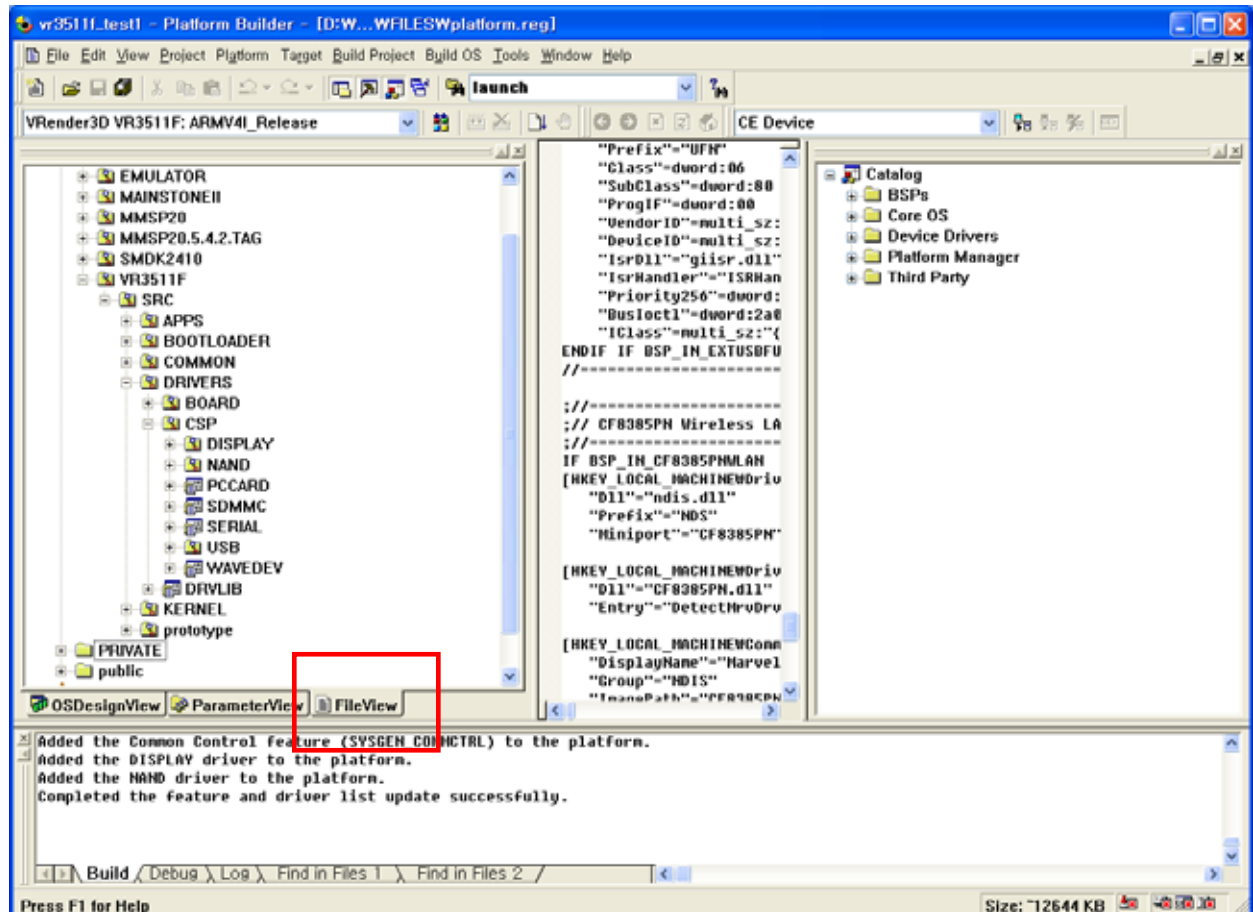
If build is done without error, download new OS image to DTK board. (See next chapter.)

15.2 Building Part of BSP Source

When you develop your own Windows CE OS image, you can build part of BSP source which you modified. By this partial building you can reduce build time dramatically.

Caution: This build method should be applied only after building entire OS is done..

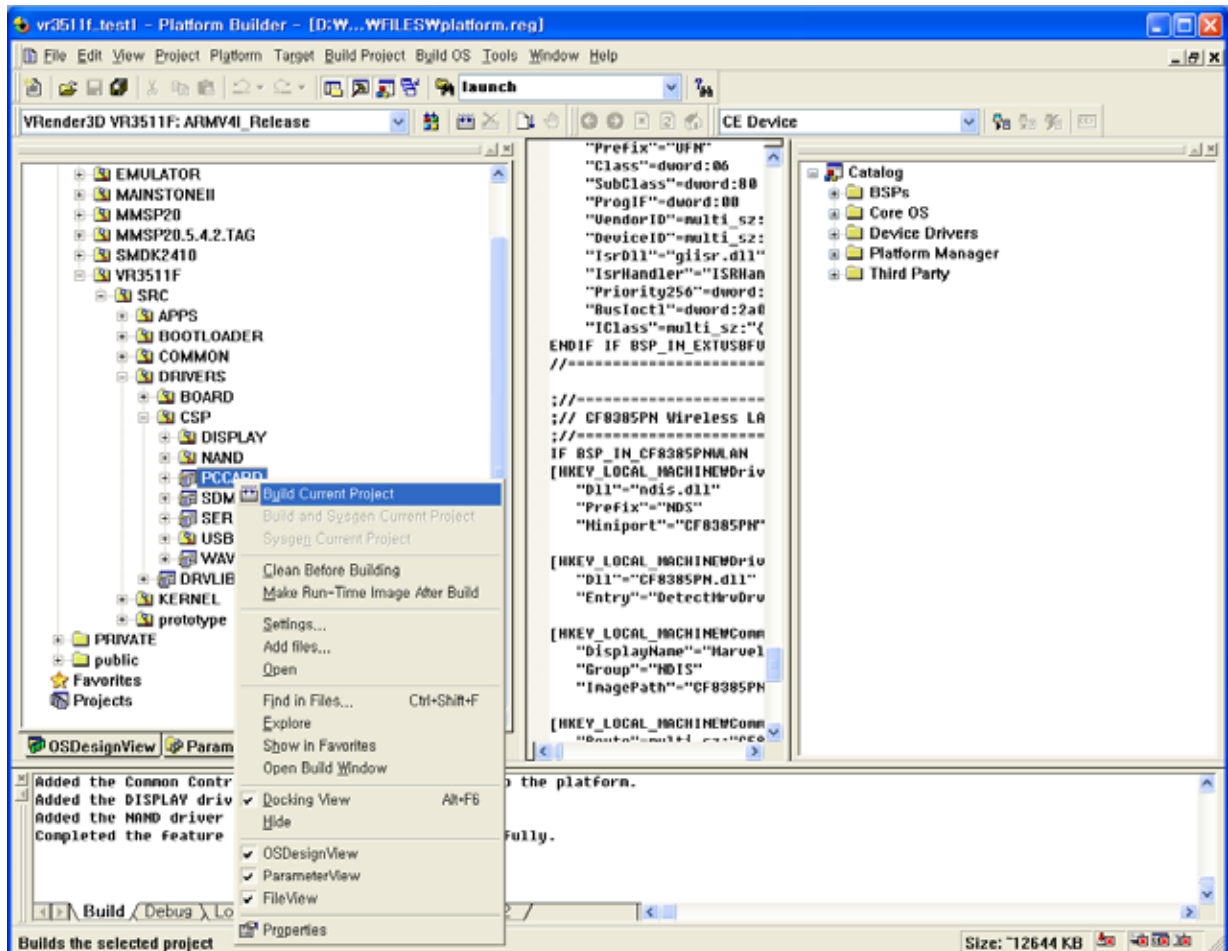
Select **File View** tab.



Select a folder that you want to build and press right button of the mouse.

When context menu is shown, select **Build Current Project**.

If **Make Run-time Image After Build** option is selected, OS image is generated after selected part of source is built and applied.



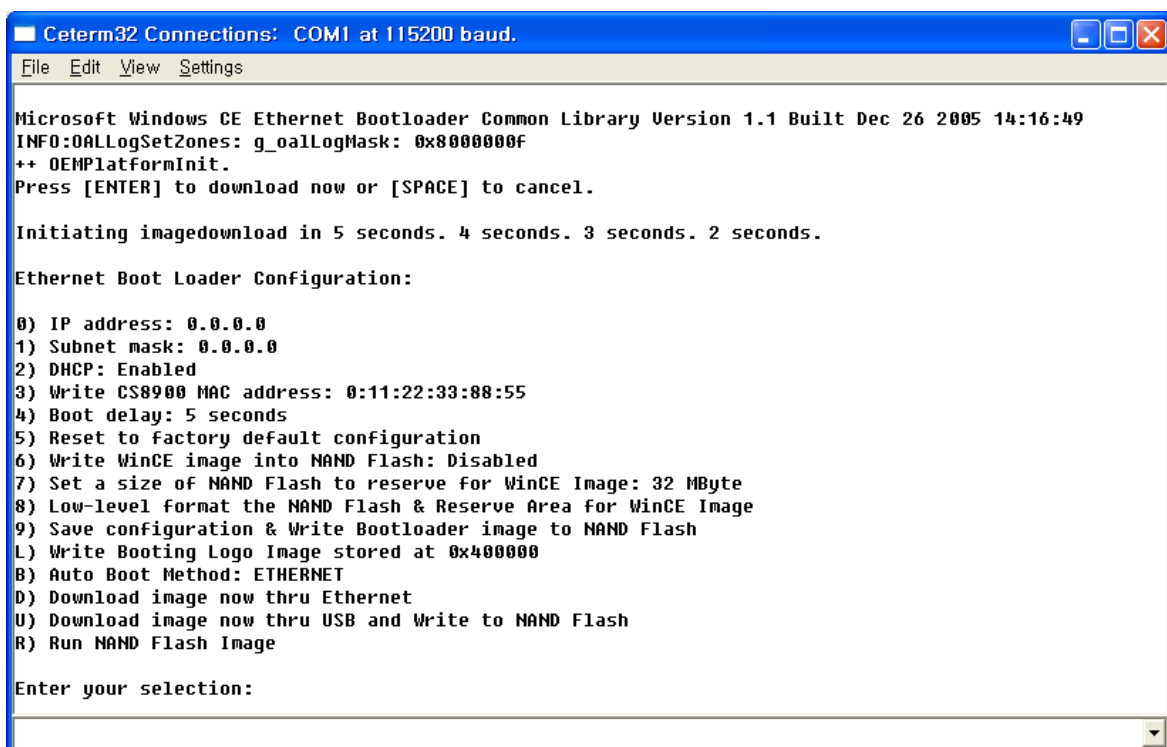
Above picture shows the process of building PCCARD driver. After building is done without error, new driver DLL file is automatically copied to release directory and making OS image is performed.

16. Download Image with Ethernet Bootloader

To download Windows CE OS image, make sure ethernet bootloader is correctly installed on DTK board.

16.1 Setting Configuration of Ethernet Bootloader

- Connect PC serial port and debugging serial port of the DTK board with cross cable.
(Debugging port is right of the two serial ports.)
- Connect LAN Cable on the board.
- Run ceterm and set baud rate to 115200bps.
 - How to run ceterm
 - 1) Run "Platform Builder".
 - 2) Select **Build OS** → **Open Release Directory** in Platform Builder Menu.
 - 3) Put "**ceterm**" on a command line.
- When connection and setting is done correctly and power of the DTK board is on, bootloader message is displayed on terminal screen. (Hyper Teminal or Ceterm)



```
Ceterm32 Connections: COM1 at 115200 baud.
File Edit View Settings

Microsoft Windows CE Ethernet Bootloader Common Library Version 1.1 Built Dec 26 2005 14:16:49
INFO:OALLogSetZones: g_oalLogMask: 0x8000000f
++ OEMPlatformInit.
Press [ENTER] to download now or [SPACE] to cancel.

Initiating imagedownload in 5 seconds. 4 seconds. 3 seconds. 2 seconds.

Ethernet Boot Loader Configuration:

0) IP address: 0.0.0.0
1) Subnet mask: 0.0.0.0
2) DHCP: Enabled
3) Write CS8900 MAC address: 0:11:22:33:88:55
4) Boot delay: 5 seconds
5) Reset to factory default configuration
6) Write WinCE image into NAND Flash: Disabled
7) Set a size of NAND Flash to reserve for WinCE Image: 32 MByte
8) Low-level format the NAND Flash & Reserve Area for WinCE Image
9) Save configuration & Write Bootloader image to NAND Flash
L) Write Booting Logo Image stored at 0x400000
B) Auto Boot Method: ETHERNET
D) Download image now thru Ethernet
U) Download image now thru USB and Write to NAND Flash
R) Run NAND Flash Image

Enter your selection:
```

16.2 Ethernet Boot Loader Configuration

Press space key when “Press[ENTER] to download now or [SPACE]to cancel.” message is shown.

16.2.1 IP address

When you set IP address, press “0” and put your IP address. An initiate value is 0.0.0.0 and you should set available IP address at your first running. Input format is xxx.xxx.xxx.xxx (e.g. 220.78.29.215).

16.2.2 Subnet mask

When you set Subnet mask, press “1” and put appropriate subnet mask. An initiate value is 0.0.0.0 and you should set available IP address at your first running. Input format is xxx.xxx.xxx.xxx (e.g. 255.255.255.0)

16.2.3 DHCP

If you would like to disable or enable DHCP, press “2” then the option is toggled. The initiate value is Enable.

16.2.4 Write CS8900 MAC address

Press “3” to set MAC address. Input format is hh.hh.hh.hh.hh.hh (e.g. 00.00.11.22.ab.cd)

16.2.5 Boot delay

If you would like to set boot delay time, press “5” and set time. The time unit is second. After the set time booting is started.

16.2.6 Reset To factory default configuration

This menu resets IP address, Subnet mask and DHCP to initiate values.

16.2.7 Write WinCE image into NAND Flash

If you would like to write WinCE image into NAND Flash or not, press “6” and this menu is enabled or disabled.

This menu is just for writing a WinCE image to NAND Flash. Therefore if this menu is enabled, booting does not take place automatically and you have to choose menu “D” to download an image with setting **Auto Boot Method** as “NAND”. When downloading is finished, “INFO : REBOOT Now” message appears then reboot the board.

16.2.8 Set a size of NAND Flash to reserve for WinCE Image

If you would like to set a size of NAND Flash for WinCE Image, press “7” and put a number for the size. This size of area is finally reserved after performing “Low-level format the NAND Flash & Reserve Area for WinCE Image” menu. 32Mbytes are recommended.

16.2.9 Low-level format the NAND Flash & Reserve Area for WinCE Image

If you would like to format the NAND Flash and reserve an area for WinCE Image, press "8". A size of the area is determined by menu "7" above. This menu should be performed at least once at the very first time.

NOTE : If you perform this menu after writing a booting logo image, the image is removed.

16.2.10 Save configuration & Write Bootloader image to NAND Flash

When you finished to set configuration, press "9" to save the configuration and the bootloader image is written to NAND Flash.

16.2.11 Write Booting Logo Image stored at 0x400000

If you would like to write booting logo image stored at 0x400000 to NAND Flash, press "L".

NOTE : Select this menu only once after downloading a logo image though ICE. If you select the menu after NAND booting, an abnormal image is saved.

16.2.12 Auto Boot Method

If you would like to choose NAND boot mode or ETHERNET boot mode while booting, press "B" then NAND or ETHERNET is selected. You need to perform menu "9" after setting to apply it.

16.2.13 Download image now thru Ethernet

If you would like to download WinCE image thru Ethernet, press "D" then downloading is started and start WinCE with the image.

16.2.14 Download image now thru USB and Write to NAND Flash

If you would like to download WinCE image thru USB, press "U" then downloading is started and the image is written on NAND Flash.

16.2.15 Run NAND Flash Image

If you would like to WinCE image stored in NAND Flash, press "R".

NOTE : It is recommended to choose menu number "7" -> "8" -> "9" -> "L" after downloading images through ICE. You are not supposed to select menu "7", "8", "L" after NAND booting. If you do so, Booting Logo Image can be damaged.

Setting Network Options to Download WinCE image thru Ethernet

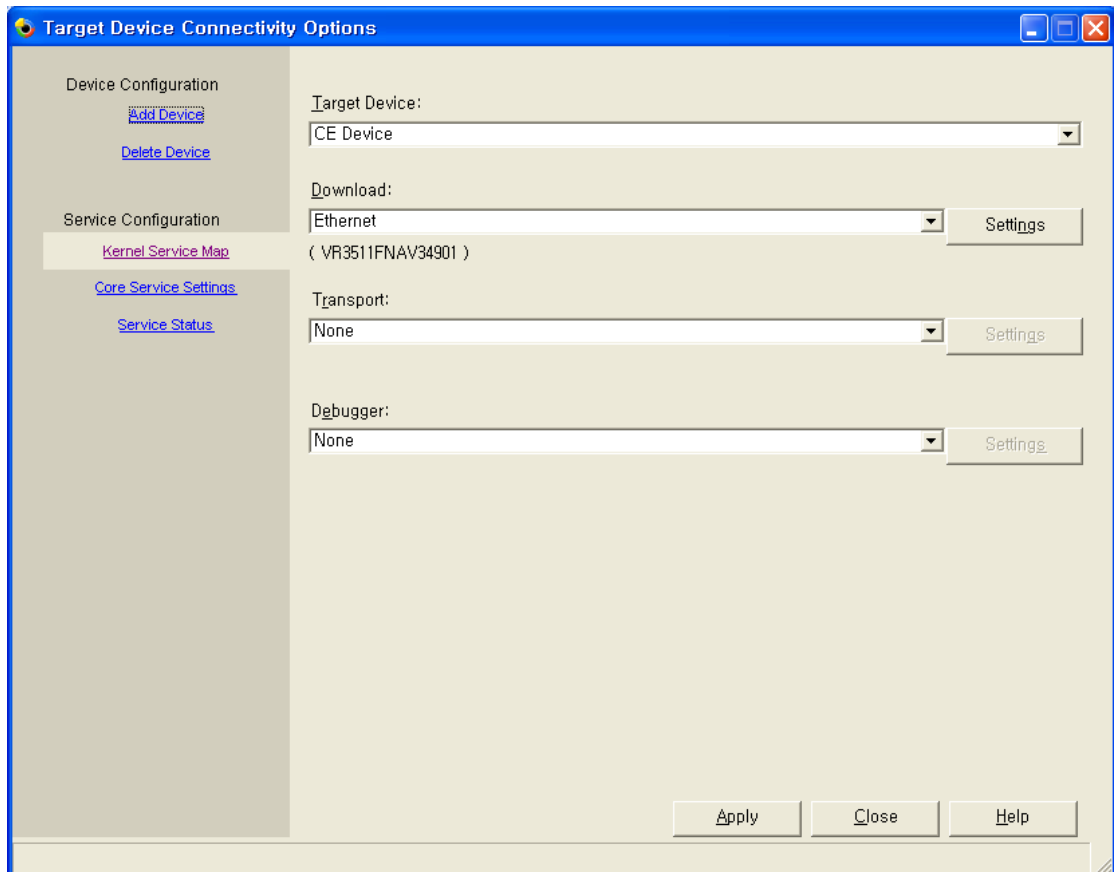
Set IP and MAC address with Ethernet Boot Loader Configuration.

After Finishing Configuration Setup, save options by selecting **Save configuration & Write Bootloader image to NAND Flash** and select **Download image now thru Ethernet**. Now bootloader will be run according to the stored options.

Setting Platform Builder for OS Image Download

Select **Target** -> **Connectivity Options** in menu.

Set **Download** as 'Ethernet' and prepare to select download target by pressing 'Settings' button.



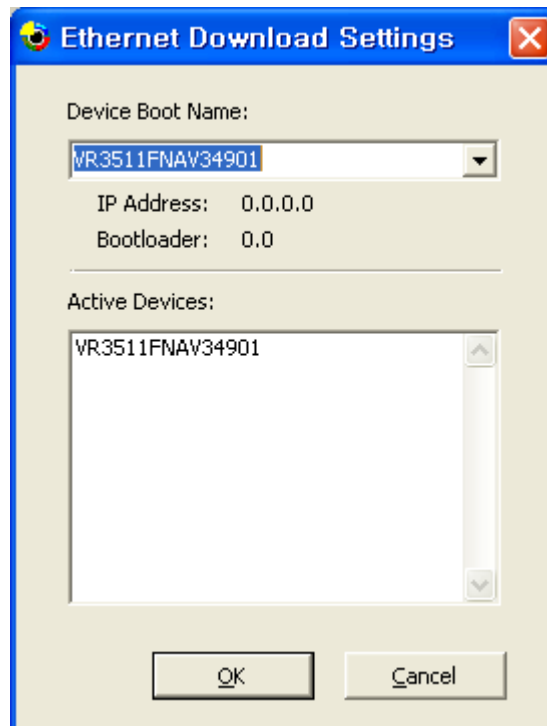
Bootloader sends BOOTME packet when boot mode is set as Ethernet download mode.

```
Cterm32 Connections: COM1 at 115200 baud.
File Edit View Settings
Microsoft Windows CE Ethernet Bootloader Common Library Version 1.1 Built Dec 26 2005 14:16:49
INFO:OALLogSetZones: g_oallogMask: 0x8000000f
++ OEMPlatformInit.
Press [ENTER] to download now or [SPACE] to cancel.

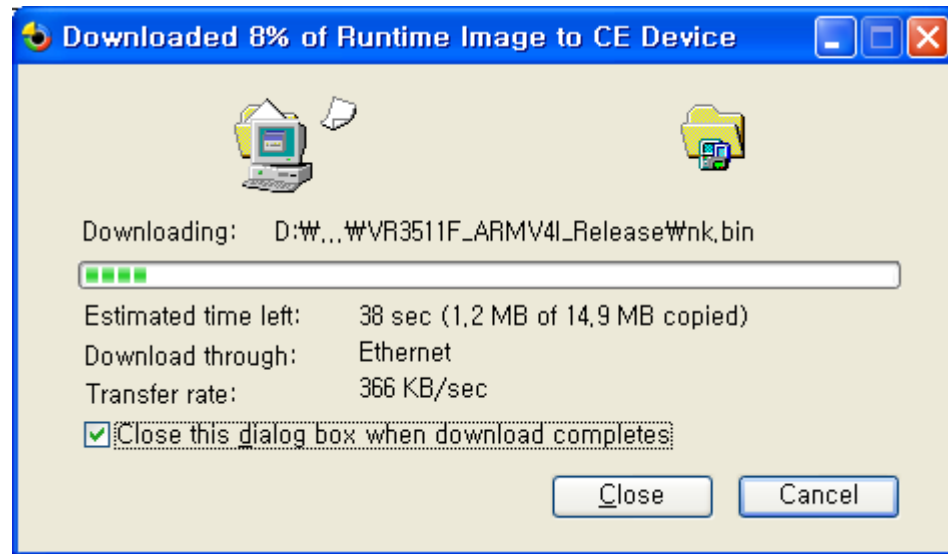
Initiating imagedownload in 5 seconds. 4 seconds. 3 seconds. 2 seconds. 1 seconds. 0 seconds.

Starting auto-download ...
+InitEthDevice.
INFO : Ethernet : UA[b4400000], PA[8c000000]
INFO: CS8900AInit chip detected
-InitEthDevice.
-- OEMPlatformInit.
System ready!
Preparing for download...
++ OEMPreDownload
+OALKit1CreateName('VR3511FNAUI', 0xa0080030, 0xa0080008)
-OALKit1CreateName(pBuffer = 'VR3511FNAV34901')
INFO: *** Device Name 'VR3511FNAV34901' ***
INFO: Using static IP address 207.924.
INFO: Using subnet mask 25252518.
+EbootSendBootmeAndWaitForTftp
Sent BOOTME to 255.255.255.255
Sent BOOTME to 255.255.255.255
Sent BOOTME to 255.255.255.255
Sent BOOTME to 255.255.255.255
Sent BOOTME to 255.255.255.255
Sent BOOTME to 255.255.255.255
```

Device name of the target board is shown on the **Active Devices** list when Platform builder receives BOOTME packet from the board. Select one from the list.



Finish configuration and run Ethernet download by selecting **Target -> Attach Device**. (Starting download is possible only when bootloader sends BOOTME packet.)



After downloading is completed, OS image runs. If you selected option that store received image to flash memory, it takes some time to store image and run OS.

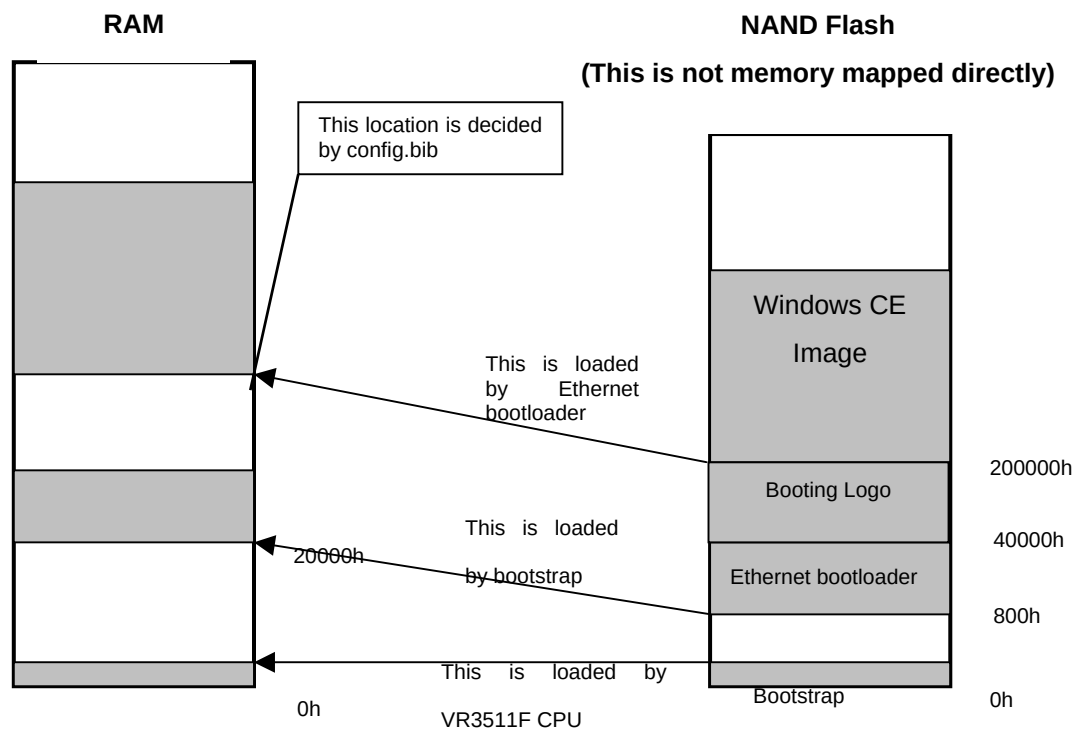
17. Installing Bootloader Image to NAND Flash

Bootloader for Windows CE image is installed when DTK board is delivered. You can replace this bootloader image with ARM debugging tools and DTK diagnostic program.

17.1 Boot Sequence

Current version of bootloader used on VR3511F DTK board is divided with two steps of images. First bootlaoder is a small image (less than 512 bytes) and called 'bootstrap.' This bootstrap is loaded by VR3511F CPU and it launches second bootloader. This second bootloader is Ethernet bootloader that you have seen in previous chapter.

- A. When power is on, VR3511F CPU loads bootstrap stored in address 0h of NAND flash. This bootstrap image is located in address 0h of the RAM.
- B. As bootstrap runs, it loads Ethernet bootloader image stored in 800h of NAND flash. This Ethernet bootloader image is located in address 20000h of the RAM.
- C. Ethernet bootloader loads Windows CE image from NAND flash or get new image through Ethernet.



17.2 Boot Sequence

Because address of NAND flash is not memory mapped directly, a program that accesses to NAND flash interface is needed to write images in NAND flash.

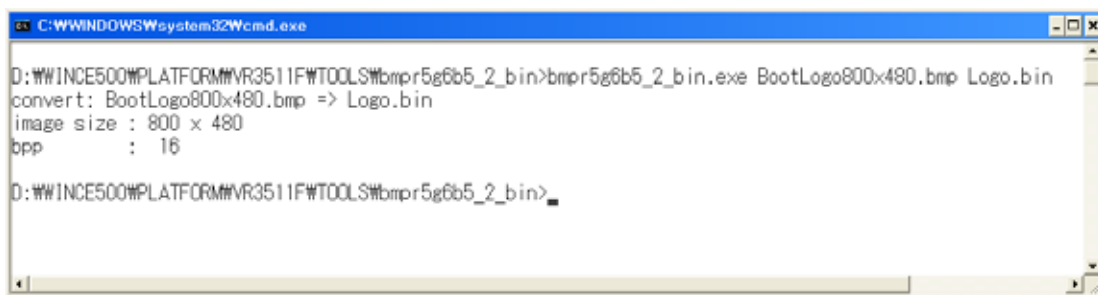
17.2.1 Requirements

To install or replace bootloader image, you must have:

17.2.2 Making Booting Logo

Run a command shell. Go to `$(_WINCEROOT)\PLATFORM\VR3511F\TOOLS\bmpr5g6b5_2_bin` folder.

Run command like this:



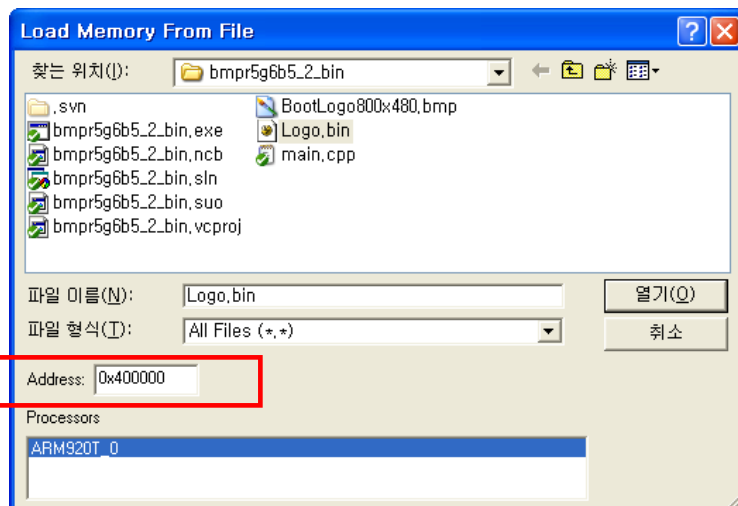
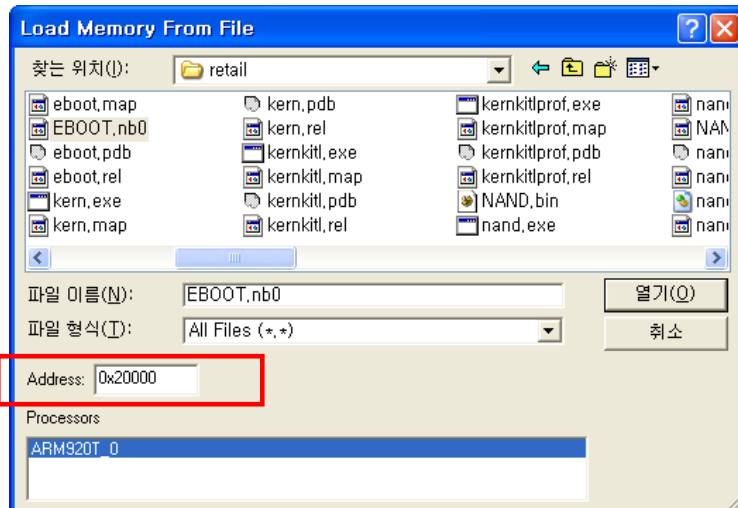
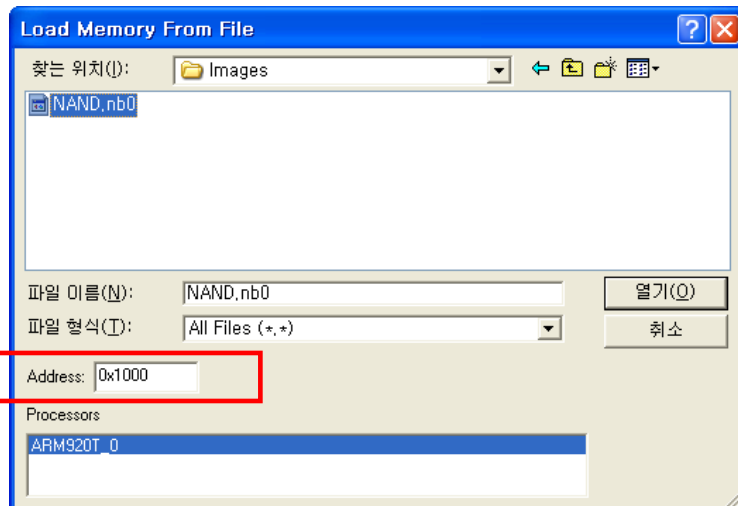
```
C:\WINDOWS\system32\cmd.exe
D:\WINCE500\PLATFORM\VR3511F\TOOLS\bmpr5g6b5_2_bin>bmpr5g6b5_2_bin.exe BootLogo800x480.bmp Logo.bin
convert: BootLogo800x480.bmp => Logo.bin
image size : 800 x 480
bpp        : 16
D:\WINCE500\PLATFORM\VR3511F\TOOLS\bmpr5g6b5_2_bin>
```

The bitmap image size has to be the same as LCD and 16-bit color format is required.

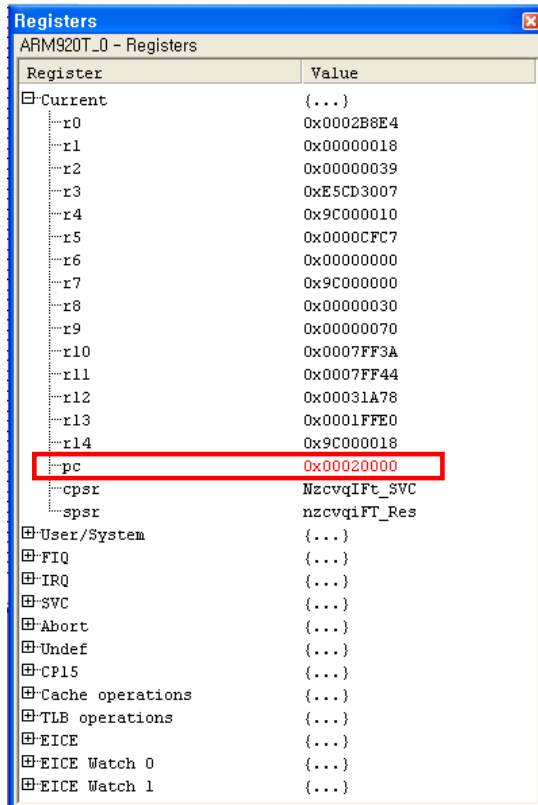
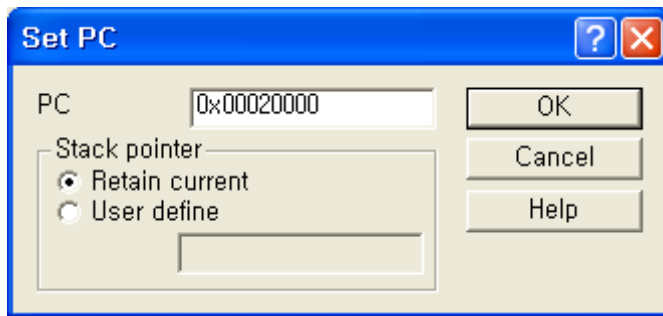
17.2.3 Installing bootloader images

Write NAND Boot Code, EBoot Code and Logo Image file using Multi-ICE or Openice.

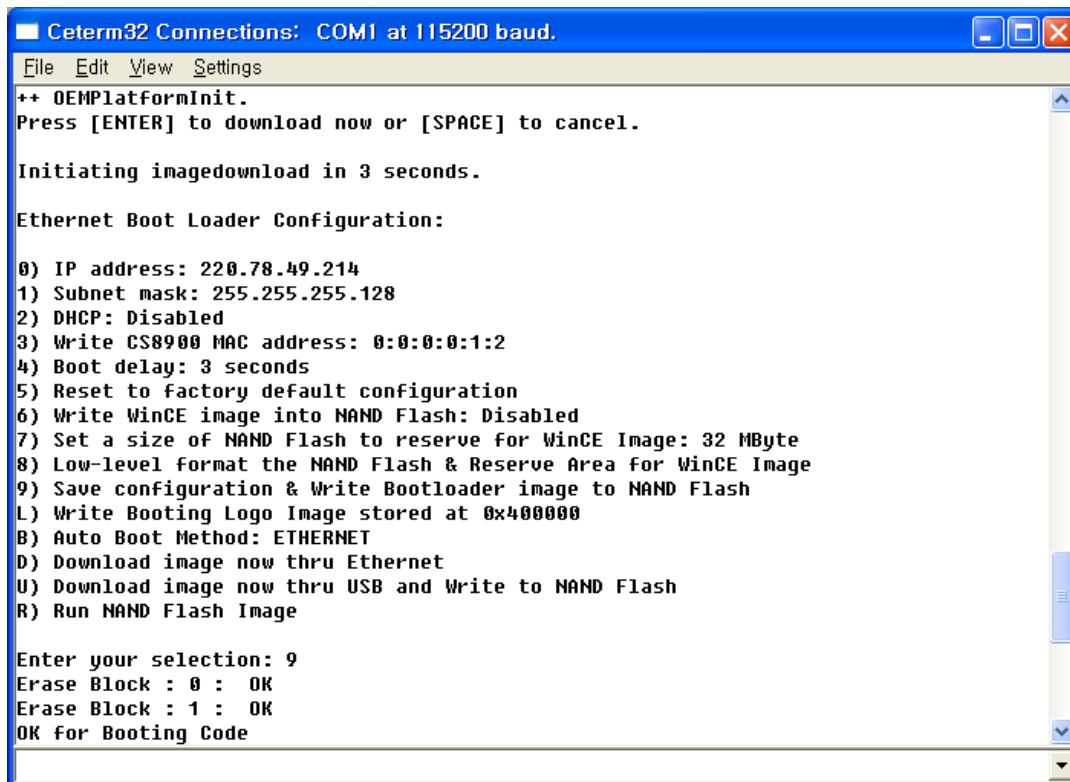
- A. Connect Multi-ICE and Run ARM Multi-ICE Server and terminal.
- B. Turn on power of a target board and run **File** → **Auto-Configure**.
- C. Run AXD and select **File** → **Load Memory From File** to load new bootloader image to RAM.
Load NAND.nb0 (bootstrap image) to 1000h, EBOOT.nb0 (Ethernet bootloader image) to 20000h
and binary code of logo image (see chapter 17.2.2 how to make it.) to 400000h.
(You can get NAND.nb0 and EBOOT.nb0 in `$(_WINCEROOT)\PLATFORM\VR3511F\target\ARMV4\retail` if the BSP has been built)



D. Set PC value to 20000h.



- E. Select **Excute** → **Go** or press F5 to run diagnostic program.
- F. Press space bar right after step E at terminal. To save logo image to NAND, press **L** and set other configuration then press **9** to save all configuration.



A screenshot of a Ceterm32 terminal window titled "Ceterm32 Connections: COM1 at 115200 baud." The window has a menu bar with "File", "Edit", "View", and "Settings". The terminal text is as follows:

```
++ OEMPlatformInit.  
Press [ENTER] to download now or [SPACE] to cancel.  
  
Initiating imagedownload in 3 seconds.  
  
Ethernet Boot Loader Configuration:  
  
0) IP address: 220.78.49.214  
1) Subnet mask: 255.255.255.128  
2) DHCP: Disabled  
3) Write CS8900 MAC address: 0:0:0:0:1:2  
4) Boot delay: 3 seconds  
5) Reset to factory default configuration  
6) Write WinCE image into NAND Flash: Disabled  
7) Set a size of NAND Flash to reserve for WinCE Image: 32 MByte  
8) Low-level format the NAND Flash & Reserve Area for WinCE Image  
9) Save configuration & Write Bootloader image to NAND Flash  
L) Write Booting Logo Image stored at 0x400000  
B) Auto Boot Method: ETHERNET  
D) Download image now thru Ethernet  
U) Download image now thru USB and Write to NAND Flash  
R) Run NAND Flash Image  
  
Enter your selection: 9  
Erase Block : 0 : OK  
Erase Block : 1 : OK  
OK for Booting Code
```

18. Using Debugger with KITL Connection

KITL (Kernel Independent Transport Layer) is a communication layer that manages communication between the development workstation and the target device. You can use KITL for kernel debugging and platform debugging.

This BSP supports KITL link with CS8900 Ethernet controller.

Note:

- You should build OS with **Debug** mode for source level debugging.
- Size of Debug mode OS image is much bigger than image with Release mode. Make sure OS image size is less than 30MB. To reduce OS image size, remove large size components such as internet explorer.
- You cannot use KITL with Ethernet and CS8900 Network driver simultaneously. Make sure CS8900 Network driver is not added from catalog.

18.1 Building CS8900 Ethernet Debugging Library

You must modify a source of CS8900 Ethernet debugging library. Location of the source file of this library is

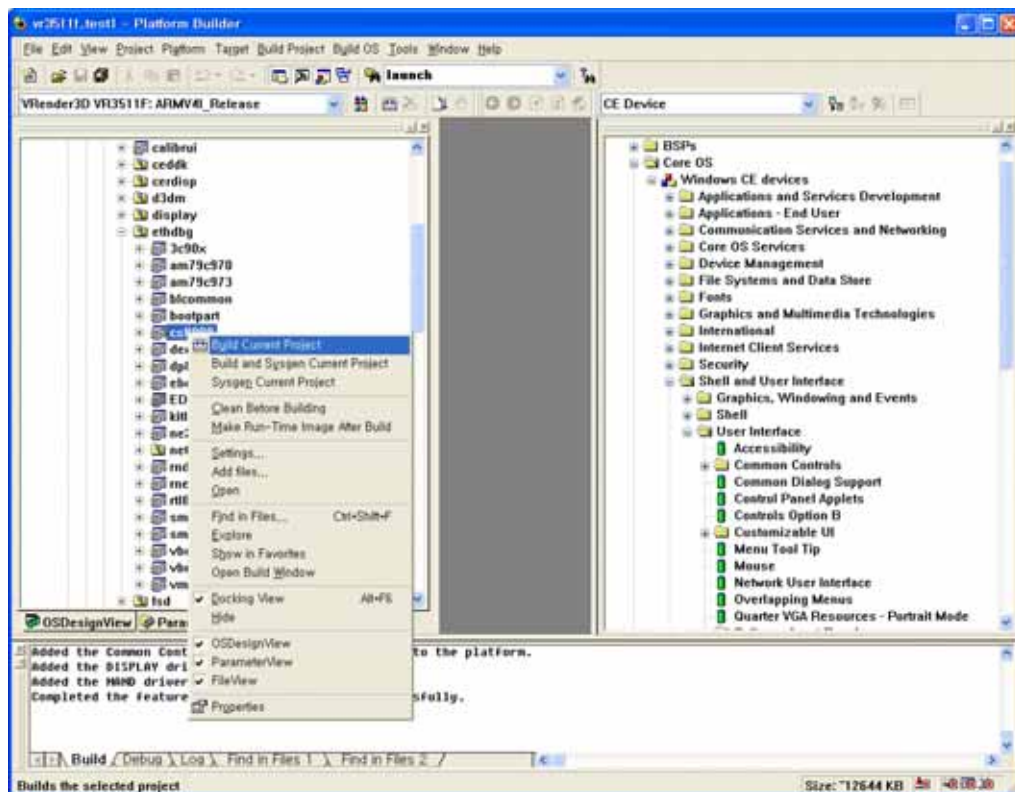
\\WINCE500\\PUBLIC\\COMMON\\OAK\\DRIVERS\\ETHDBG\\CS8900.

Open **cs8900dbg.c** file.

Find the line shown below. Remove this line or cover it with a comment

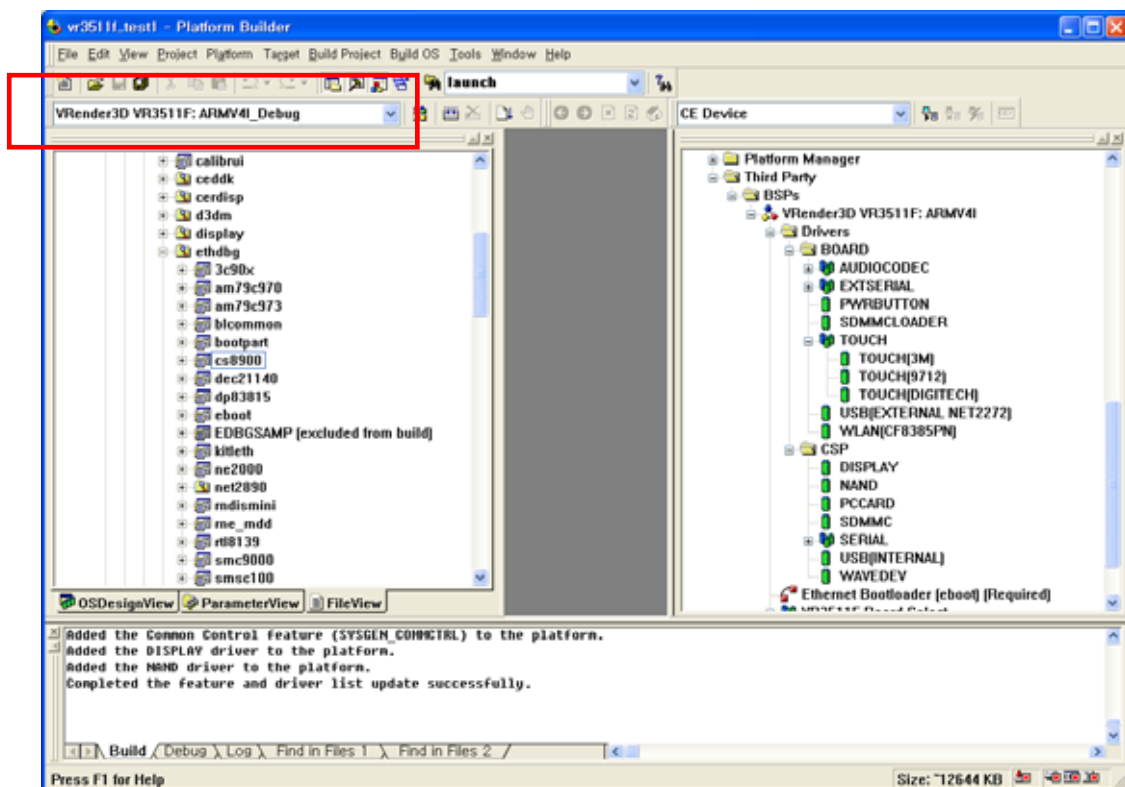
#define CS8900_MEM_MODE

Build this library. Select CS8900 directory on **FileView** and click it with right button of the mouse. Then perform building by selecting **Build Current Project**. (For detailed information of building part of source code, see **Building WinCE OS -> Building Part of BSP Source** chapter of this document.)



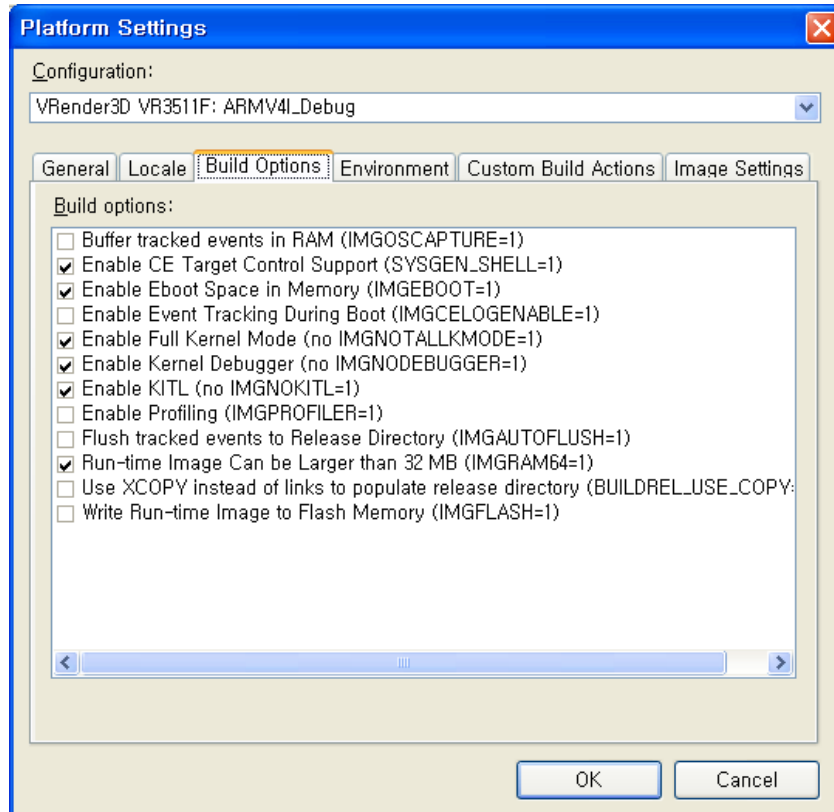
18.2 Setting and Building OS Image for KITL Link

For source level debugging, OS image should be built Debug mode.

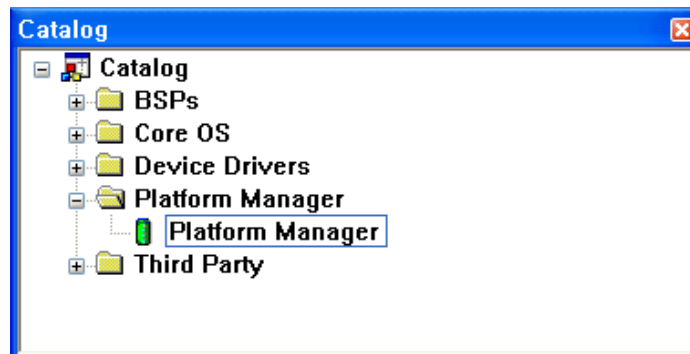


Select **Platform** -> **Settings** menu and select **Build Options** tab.

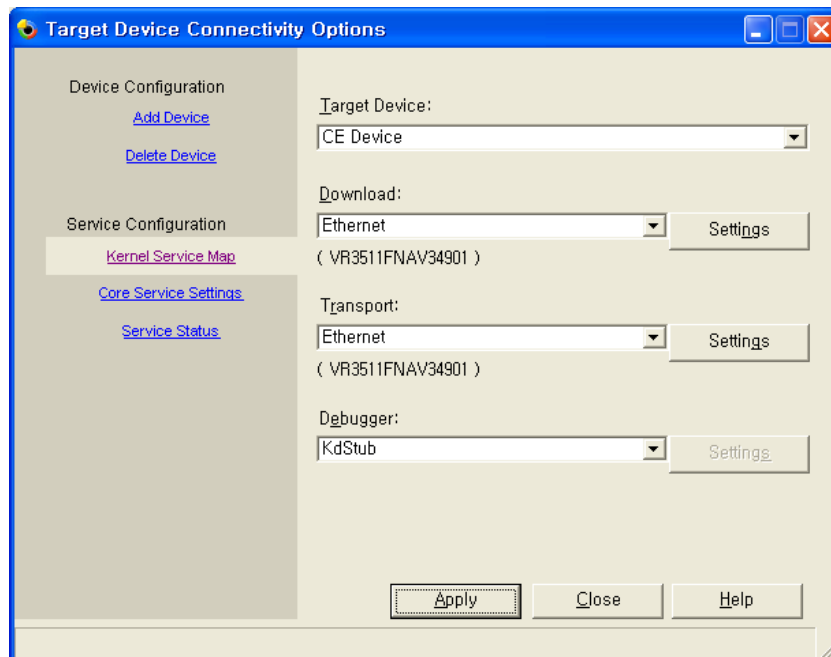
Select **Enable Kernel Debugger** and **Enable KITL** options.



Add **Platform Manager** -> **Platform Manager** from **Catalog** list.



Select **Target** -> **Connectivity Options** and select **Debugger** of current connectivity selection as **Ethernet**.



Build OS by selecting **Build OS -> Sysgen**.

18.3 Connect Platform Builder and Target System with KITL

Download OS image with Ethernet bootloader.

After download is completed, you can see this debugging window. Now you can use kernel debugging.

